



Advanced machine learning models for accurate water quality classification and WQI prediction: Implications for aquatic disease risk management

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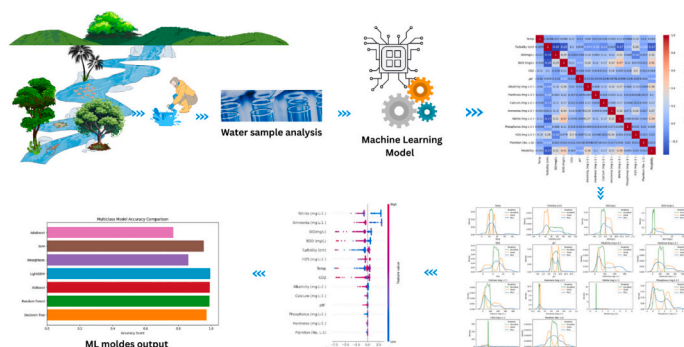
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HIGHLIGHTS

- LightGBM & XGBoost hit 99.65 % accuracy, R^2 up to 0.97 for WQI prediction.
- DO, BOD, turbidity, nitrate & EC are key indicators of aquatic disease risk.
- Hyperparameter tuning greatly boosted model performance in all tasks.
- ML with real-time water data enables early warnings for aquatic health.

GRAPHICAL ABSTRACT



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ABSTRACT

Accurate classification of water quality and precise prediction of the Water Quality Index (WQI) are essential for safeguarding aquatic ecosystems and mitigating disease risks in aquaculture. This study systematically evaluates multiple machine learning models including LightGBM, XGBoost, Random Forest, and Support Vector Machines using grid search optimization to enhance performance. Results show that ensemble models, particularly LightGBM and XGBoost, achieved superior accuracy in water quality classification (up to 99.65 %). For WQI prediction, XGBoost regression delivered the highest performance with an R^2 of 0.9685. SHAP (SHapley Additive exPlanations) analysis was employed to interpret model predictions and quantify feature contributions. Dissolved oxygen and BOD emerged as the most influential predictors, followed by turbidity, nitrate, and electrical

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conductivity, aligning with known risk factors for aquatic disease outbreaks. These findings underscore the potential of combining advanced machine learning with explainable AI techniques and real-time water quality data to enable proactive monitoring and early warning systems for sustainable aquatic health management.

1. Introduction

Water is one of the most vital resources for the survival of all living organisms. However, contamination can degrade water quality, increasing the prevalence of aquatic diseases that may affect human health through the consumption of infected or bioaccumulated organisms (Kılıç, 2020). As a result, monitoring water quality and ensuring the protection and longevity of marine biodiversity is crucial. Moreover, addressing water quality issues is essential for the management and mitigation of pollution (Kumar et al., 2022). Numerous countries internationally are implementing ecological water management initiatives to evaluate the health of aquatic ecosystems. At present, approximately one billion individuals do not have access to safe drinking water, resulting in around two million deaths each year from contaminated water and insufficient sanitation (Jain, 2012; Gleick, 2002). Thus, preserving the quality of freshwater is essential. Maintaining good water quality is a prerequisite for sustainability of any water diversion project. Poor water quality might result in significant costs, as investment must be allocated in resources in the form of routine maintenance or fixing the water supply systems whenever problems arise. Subsequently, there is a growing need for better water management and quality control measures to secure safe drinking water at a reasonable price (Yu et al., 2018). Addressing these challenges necessitates comprehensive risk assessments of freshwater supplies, waste management systems, and institutional surveillance mechanisms. Water quality prediction involves evaluating and forecasting the condition of a water ecosystem at a given point in time, based on various physicochemical and biological parameters (Wang et al., 2016). This approach has implications for the planning and management of water resources. Through predicting future water quality trends of existing pollution patterns, more effective policies for pollution prevention and control would be facilitated (Dong et al., 2013). In water diversion projects, the overall water quality must be assessed, especially when large volumes of water are transported daily for drinking purposes. Consequently, it has become more valuable to have predictive methodologies with respect to water quality in the current world. Advancements in artificial intelligence (AI) and machine learning (ML) are crucial for addressing the security issues. These tools emphasize the identification of relationships between inputs and outcomes in water systems, moving beyond complex procedural models.

Water quality prediction is important in water resource planning, management, and control. It is a main aspect of research to monitor environmental water quality to preserve the aquatic ecosystem. Therefore, it is crucial to establish a practical and trustworthy method for predicting water quality. Meanwhile, anticipating future changes in water quality is vital for mitigating unanticipated variations and performing suitable interventions. Accurate forecasting of water quality variations ensures drinking water safety and supports sustainable fisheries and the protection of aquatic biodiversity (Wu and Wang, 2022). Moreover, traditional methods of predicting water quality frequently fail to account sufficiently for the impacts of biological, physical, hydraulic, geochemical, and meteorological factors. Researchers are currently focusing on enhancing the reliability and applicability of groundwater prediction models. To overcome current limitations, several advanced methodologies have been proposed, including artificial neural networks (ANN), stochastic mathematics, fuzzy logic approaches, and geospatial technologies (3S), aimed at improving forecasting accuracy and expanding their practical applications (Pan et al., 2025).

The incorporation of technologies, including remote sensing, cloud computing, the IoT, big data, and artificial intelligence, has substantially

improved the development and implementation of aquatic environment monitoring systems. Smart monitoring techniques have been implemented in various areas of China, employing diverse infrastructure such as automated hydrological and water quality monitoring stations, wireless sensor networks (WSNs), remote sensing-based systems, monitoring vessels, and advanced underwater robotic devices (Liu et al., 2019). Artificial intelligence holds significant potential for improving the efficiency of water supply and sanitation operations, while also ensuring compliance with standards for drinking water and wastewater treatment. Consequently, significant scientific focus has been allocated to the modeling and forecasting of water quality to control and mitigate pollution efficiently (Hmoud Al-Adhaileh et al., 2021).

The WQI serves as a standard metric for evaluating overall water quality in diverse contexts, which include drinking water, industrial applications, and the health of aquatic ecosystems. A higher WQI value generally indicates better water quality. The Water Quality Classification (WQC) system classifies water bodies as clean or slightly polluted based on WQI ranges. The WQI consolidates various water quality parameters assessed at specific locations and times. However, calculating the WQI requires sub-index computations, which can be both time-intensive and prone to errors. Thus, establishing precise and effective WQI prediction methodologies is extremely important (Abbas et al., 2024).

Nonlinear relationships in the studied system can be modelled effectively using extensive historical data that captures dynamic development operation, regardless of previous domain knowledge. Access to clean water is essential for all life forms; hence, developing reliable forecasting methods for predicting future water quality conditions is of considerable social and economic significance. In recent decades, pollution has significantly impacted water quality, negatively affecting both aquatic ecosystems and public health (Wu and Wang, 2022).

Evaluating and comprehending water quality is fundamental for the sustainable management of water resources. The WQI serves as a recognized tool that integrates various physical, chemical, and biological parameters, including pH, dissolved oxygen, turbidity, nutrient concentrations, and pollutants into a single, measurable value. This provides data visualization and interpretation which is easily comparable across locations and periods. Integrating these variables allows WQI to support informed decision-making in water resource management. WQC categorises water depending upon threshold values within a certain range of quality levels, which provides a practical way to indicate the pollution status. This classification assists stakeholders in identifying key areas, prioritising remediation efforts, and enacting suitable regulatory measures to safeguard water quality (Shams et al., 2023).

The current study is motivated to contribute to addressing the deterioration of water quality and its impacts on the ecosystems, public health, and sustainable development. Water quality monitoring and evaluation is essential in understanding potential threats, enacting appropriate management policies and realizing the goal to provide clean water. Conventional methods for the water quality analysis including laboratory tests and the calculation of the WQI based on measured parameters are time-consuming and expensive and limited in delivering real-time data. Conversely, predictive modeling through machine learning provides a convenient alternative to estimating WQI and WQC based on the existing data. These models enable timely measurement of water quality, even in situations when direct measurement of any parameter is not possible.

The application of machine learning for predicting WQI and WQC facilitates timely monitoring, providing near-real-time estimates that

improve efficiency, lower costs, and enable early detection of pollution or quality deterioration. Moreover, these models are capable of handling incomplete data sets during the estimation of missing values, indicating a robust WQI calculation even with certain parameters are unavailable. This approach significantly enhances the monitoring and management of water resources.

However, to fully capture both the categorical status of water bodies and the exact pollutant levels, it is necessary to integrate classification and regression approaches within a single framework.

While regression models WQI provide continuous and precise quantification of WQC enable rapid grading into categories such as clean, moderately polluted, or highly polluted. Combining these two approaches provides a synergistic framework: classification supports fast decision-making and early warning, while regression ensures detailed quantification for long-term management and compliance. This dual-task perspective strengthens the practical utility of machine learning in water quality monitoring, making the predictions both actionable and precise.

Accurate estimation of WQI and WQC allows optimized allocation of resources by providing better direction of sampling, monitoring, and treatment based on the expected quality of the water (Al Mamun Hridoy et al., 2025). Predictive models facilitate the creation of early warning systems that continually predict water quality trends and enable timely preventive actions to avoid deterioration and protect water resources. This study adopts a dual-task approach, employing machine learning algorithms to forecast both WQC (classification) for rapid water quality grading and WQI (regression) for precise pollution quantification. Parameter tuning is performed with grid search over four classification models: random forest (RF), extreme gradient boosting (XGBoost), gradient boosting (GB) and adaptive boosting (AdaBoost) for WQC prediction and four regression models - K-nearest neighbors (KNN), DT, support vector regressor (SVR), and multi-layer perceptron (MLP) for WQI prediction. In classification, experimental results indicated that gradient boosting-based models (LightGBM and XGBoost) provided the highest accuracy (99.65 %) for WQC, and MLP regressor outperforms others, with an R^2 of 99.8 %, for WQI prediction. The contributions of this paper are outlined as follows:

- Systematic benchmarking of ML models: We conduct a rigorous, leak-safe comparison of multiple machine learning models for both WQI regression and water quality classification, ensuring fair evaluation under a unified pipeline.
- Comprehensive hyperparameter optimization: We apply grid search with nested cross-validation to optimize model parameters, reducing bias and improving reliability in both regression and classification tasks.
- Multi-metric performance evaluation: Beyond accuracy, we report micro/macro/weighted F1, MCC, and calibration analysis, providing a holistic view of model performance and robustness under class imbalance.
- Practical deployment insights: We analyze computational efficiency, interpretability, and robustness, offering guidelines for real-world aquaculture monitoring systems aimed at mitigating aquatic disease risks.
- Open-source reproducibility: All code, tuned configurations, and evaluation scripts are made publicly available to facilitate replication and future research.

2. Related work

SVR, grey system (GS), and regression analysis (RA) have been widely adopted for water quality prediction in recent years. Numerous studies have explored advanced machine learning models in this domain. Such, Liu et al. (2019) employed a long short-term memory (LSTM) network to forecast drinking water quality in the Yangtze River Basin based on variables such as COD, DO, pH, and $\text{NH}_3\text{-N}$,

demonstrating the model's effectiveness for real-time monitoring. Similarly, Khullar and Singh (2021) developed a Bi-LSTM-based deep learning model (DLBL-WQA) for the Yamuna River, reporting notable improvements in accuracy and error metrics compared to traditional models. Similarly, developed a hybrid CEEMDAN-BiGRU-SVR-MWOA framework for lake water level prediction, demonstrating how multi-source fusion and decomposition-reconstruction strategies can significantly enhance time-series forecasting accuracy. Abba et al. (2020) evaluated multiple models—ANFIS, BPNN, MLP, and SVR—for WQI prediction and identified the Neural Network Ensemble (NNE) as the best performer across three monitoring stations. In addition, Elbeltagi et al. (2021) compared four streamflow simulation tools (M5P, AR, Support Vector Machine (SVM), RSS and found that the AR model achieved the highest accuracy and stability using fewer input variables. Likewise, Asadollah et al. (2021) applied an Extra Tree Regression (ETR) ensemble method to predict monthly WQI in Hong Kong, outperforming conventional standalone models in both efficiency and accuracy.

Furthermore, Nosair et al. (2022) used AI-integrated regression models with Salinity Water Index (SWI) indicators to predict salinization risks in the Nile Delta aquifer, while Alsulamy et al. (2025) employed ensemble techniques—such as Random Forest, XGBoost, LogitBoost, and AdaBoost—for water classification, with XGBoost achieving a peak accuracy of 96.97 % (Table 1). In a broader regional study, Hassan et al. (2021) applied a range of models including neural networks, random forest, SVM, and multiple linear regression to classify water quality across various regions of the Indian subcontinent, achieving extremely high accuracy levels (up to 99.99 %) and identifying key parameters such as DO, BOD, TC, nitrate, EC, and pH.

Further demonstrated how hybrid architectures (CNN-BiLSTM with attention and intelligent optimization) can effectively capture both spatial and temporal dependencies in lake water quality prediction, specifically dissolved oxygen. Their approach not only improved accuracy but also proposed a control mechanism (temperature regulation) to adjust water quality outcomes in aquaculture contexts, showing a direct link between prediction models and practical management. This highlights why recurrent models such as LSTM and Bi-LSTM are particularly suited for time-series data, and illustrates how most prior studies, unlike stop short of integrating prediction with real-time intervention strategies.

More recently, a hybrid BPNN model optimized with an Adaptive Evolutionary Artificial Bee Colony (AEABC) algorithm, achieving strong robustness even under missing or erroneous values. Their results emphasize how nature-inspired optimisation can overcome the limitations of conventional neural networks in handling noisy and nonlinear water quality data. This illustrates that while LSTMs are well-suited for sequential prediction, hybrid evolutionary-ANN approaches like AEABC-BPNN may be more effective for long-term stability and error tolerance, especially when datasets are imperfect.

Despite these advancements, several notable gaps persist in the current body of literature. Most studies are geographically or temporally constrained, limiting the generalizability of the models across diverse aquatic environments. However, much of the existing literature tends to catalogue model outcomes without critically examining why certain approaches are more (or less) suited to specific prediction tasks, for instance, the time-series strengths of LSTM models versus the robustness of hybrid optimization-based ANNs. Similarly, recurring limitations such as single-dataset validation or lack of real-time sensor integration are often mentioned in general terms but rarely backed with concrete study examples, which weakens the persuasiveness of these critiques (Rajae et al., 2019).

Moreover, many rely on datasets with limited spatial and temporal coverage or a narrow range of water quality parameters. In contrast, the present study introduces a more diverse and temporally comprehensive dataset, capturing a wide array of physicochemical and microbial indicators across multiple seasons and monitoring sites. Another limitation of earlier work lies in evaluation approaches, which often prioritize

Table 1
Machine learning techniques for water quality prediction.

Author	Technique(s) Used	Best Model	Prediction Index	Results	Dataset Characteristics	Evaluation Metrics	Application Scenario
Radhakrishnan and Pillai (2020)	SVM, Decision Tree, Naïve Bayes	Decision Tree Algorithm	Weighted Arithmetic Water Quality Index (WAWQI)	Accuracy = 98.50 %	Two datasets: (1) Narmada River (2017–2018), 28 parameters, monthly sampling; (2) Historical India dataset (2003–2014), 8 parameters, 1991 samples	Accuracy, Balanced Accuracy, Confusion Matrix	Surface water quality monitoring (river, national datasets, drinking/irrigation suitability)
Jain et al. (2021)	Random Forest Algorithm, SVM, KNN	Random Forest Algorithm	WQI	Accuracy = 92.127 %	~2000 records, Indian water quality data (2003–2014); 12 features (Temp, DO, pH, EC, BOD, nitrate/nitrite, total coliform, year, location, etc.)	Accuracy, Confusion Matrix, K-fold CV, Gini Index, Entropy, Information Gain	Marine life sustainability assessment (India, statewide rivers/lakes)
Hmoud Al-Adhaileh and Waselallah Alsaade (2021)	Neuro-Fuzzy Inference System (ANFIS), KNN, Feed-Forward Neural Network (FFNN)	ANFIS for WQI and FFNN for WQC	WQC, WQI	Accuracy (ANFIS) = 96.17 %, Accuracy (FFNN) = 100 %	1679 samples from 666 rivers/lakes in India (2005–2014); 7 key parameters (DO, pH, conductivity, BOD, nitrate, fecal coliform, total coliform)	MSE, RMSE, Mean Error, Pearson's R, Accuracy, Sensitivity, Specificity, Precision, Recall, F-score	Drinking water monitoring and contaminated water management (surface water, India)
Malek et al. (2022)	Decision Tree (DT), Naïve Bayes, Gradient Boosting, KNN, ANN, Random Forest (RF), SVM	Gradient Boosting	Water Quality Classification (WQC)	Accuracy = 94.90 %, Sensitivity = 80.0 %, F-measure = 86.49 %	658 samples (2005–2020) from Kelantan River, Malaysia; 13 parameters (DO, BOD, COD, TSS, pH, NH ₃ -N, temp, EC, salinity, turbidity, NO ₃ , PO ₄ , <i>E. coli</i>)	Balanced Accuracy, Accuracy, Classification Error, Precision, Sensitivity, Specificity, F-measure, AUC	River water monitoring and policy support (Kelantan River, Malaysia)
Khan et al. (2022)	Principal Component Regression (PCR), Gradient Boosting Classifier (GBoost)	PCR, Gradient Boosting Classifier	WQI, Water Quality Status (WQS)	Accuracy (PCR) = 95 %, Accuracy (GBoost) = 100 %, R ² = 0.932 (PCR8)	108 samples; Gulshan Lake, Dhaka (2016); 9 water quality parameters (pH, DO, COD, TDS, EC, Turbidity, Chloride, SS, Alkalinity)	R ² , Accuracy	Urban lake water (drinking/irrigation potential, surface water monitoring)
Hridoy et al. (2025a)	TFN, Linear Regression, Random Forest	TFN Model	TFN, Linear Regression, Random Forest	R ² = 0.846, RMSE = 0.047, MAE = 0.038	6-month dataset (Jan–Jun 2024) from IoT-based monitoring in tilapia ponds, Montería, Colombia; Parameters: Temperature, DO, pH, Turbidity	R ² , RMSE, MAE, p-values, Pearson correlation	R ² , RMSE, MAE, p-values, Pearson correlation
Khoi et al. (2022)	Adaptive Boosting (AdaBoost), Gradient Boosting (GBoost), Histogram-based Gradient Boosting (HGBoost), Light Gradient Boosting Machine (LGBost), XGBoost	Extreme Gradient Boosting (XGBoost)	WQI	R ² = 0.989, RMSE = 0.107	220 samples (2010–2017), 4 monitoring stations, La Buong River, Vietnam; 10 variables (T, pH, DO, BOD, COD, NH ₄ ⁺ , PO ₄ ³⁻ , TSS, turbidity, coliform)	R ² , RMS	Surface water quality assessment, pollution management (La Buong River, Vietnam)
This study	AdaBoost, SVM, KNeighbors, XGBoost, Random Forest, Decision Tree, LightGBM	LightGBM (Classification); Random Forest Regressor (Regression)	Water Quality Index (WQI) classification and prediction	LightGBM: Accuracy = 99.65 %; RF Regressor: MAE = 4.01, R ² = 0.982; Decision Tree Regressor: MAE = 4.88, R ² = 0.975	4300 records, 14 physicochemical and biological parameters (Temp, DO, BOD, pH, nutrients, H ₂ S, plankton, etc.)	Accuracy, Precision, Recall, F1, MCC (classification); MAE, R ² , MSE, MedAE (regression)	Real-time monitoring and prediction of aquaculture water quality (potability and sustainability assessment)

accuracy alone. Our study addresses this by adopting a more holistic comparative framework that includes model interpretability, feature importance (e.g., via SHAP values), and reproducibility. This allows for better transparency and practical applicability in real-world water quality management. Furthermore, by integrating insights on variable

significance and model behavior, our findings not only enhance prediction capability but also offer actionable recommendations for environmental monitoring and policy development. Consequently, the next section explores the predictive performance and classification effectiveness of selected machine learning techniques applied to this novel

and comprehensive dataset.

3. Materials and methods

The secondary dataset utilized in this study was obtained from the Mendeley Data Repository (Dataset DOI: <https://data.mendeley.com/datasets/y78ty2g293/1>) (Venkataramana et al., 2024). This dataset comprises a collection of physicochemical parameters characterizing the quality of freshwater samples. In total, it includes 170 water samples collected from different surface and groundwater sources, providing a comprehensive representation of hydrochemical variability. The dataset covers a wide concentration range for most variables, capturing natural variability and potential outlier conditions due to anthropogenic influence. Such heterogeneity is valuable for developing robust predictive models but also necessitates careful preprocessing, including missing value treatment, outlier detection, and feature scaling. However, the dataset documentation does not specify the geographical collection area (e.g., single farm, multiple regions) or the temporal coverage (e.g., specific year, seasonal variation, inclusion of extreme weather events). This lack of spatiotemporal context limits the ability to evaluate the model's generalization across diverse environmental conditions, such as tropical rainy seasons or dry periods, where water quality parameters can fluctuate significantly.

3.1. Regression and classification models for water quality assessment

This study utilized a classification-based machine learning approach to predict the WQI using a balanced dataset comprising 4300 records. Each record consisted of 15 variables, including 14 input features and one target classification label (Table 2). The input features represent essential physicochemical and biological parameters typically measured in aquaculture water quality assessments. These variables include temperature (°C), turbidity (cm), dissolved oxygen (mg/L), biochemical oxygen demand (mg/L), carbon dioxide (mg/L), pH, alkalinity (mg/L), hardness (mg/L), calcium (mg/L), ammonia (mg/L), nitrite (mg/L), phosphorus (mg/L), hydrogen sulfide (mg/L), and plankton density (No./L). In this study, no explicit sample balancing or resampling techniques were applied. The class counts (0 = 1400, 1 = 1400, 2 = 1500; total = 4300) represent the original dataset distribution, reflecting a mild natural prevalence of the “poor” water quality category. To maintain this distribution, we adopted a stratified train–test split, resulting in 938, 938, and 1005 samples for classes 0, 1, and 2 in the training set, and 462, 462, and 495 samples, respectively, in the test set. For models sensitive to class imbalance, we employed the parameter setting `class_weight = ‘balanced’` where supported. Since no resampling was performed, the question of performance differences between balanced and original samples does not apply. To enhance clarity and reproducibility, we explicitly report per-class sample counts for both training and test partitions in the revised manuscript. These included

Table 2
Characteristics of 14 water quality parameters used in this dataset.

Parameter	Unit	Role in aquaculture
Temperature	°C	Regulates metabolism and oxygen solubility
Turbidity	cm	Affects light penetration and photosynthesis
DO	mg/L	Essential for respiration in fish and aquatic organisms
BOD	mg/L	Indicates organic load; high BOD reduces DO
CO ₂	mg/L	Excess CO ₂ can reduce pH and affect fish respiration
pH	–	Influences biochemical processes and fish health
Alkalinity	mg/L	Buffering capacity of water
Hardness	mg/L	Influences osmoregulation and shell formation
Calcium	mg/L	Crucial for bone and shell development
Ammonia	mg/L	Toxic at elevated levels; affects gill function
Nitrite	mg/L	Converts to toxic nitrate at high levels
Phosphorus	mg/L	Nutrient: high levels cause eutrophication
H ₂ S	mg/L	Highly toxic; affects respiration
Plankton	No./L	Indicates productivity and food availability

Random Forest (RF), SVM, KNN, Decision Tree (DT). The dataset was split into training and testing subsets in a 67:33 ratio, and k-fold cross-validation (with $k = 5$) was used to enhance model generalizability and reduce overfitting risks.

Model performance was evaluated using standard classification metrics, including accuracy, precision, recall, F1-score, and confusion matrix analysis (Fig. 1). Hyperparameter tuning was conducted using grid search and randomized search approaches to optimize model configurations. The structured and comprehensive nature of the dataset provided a solid foundation for building artificial intelligence models capable of assessing and predicting water quality in aquaculture systems with high reliability.

3.1.1. Decision tree (DT)

The Decision Tree Classifier works by recursively splitting the dataset based on the feature that results in the highest information gain, constructing a tree structure where each internal node represents a decision rule, and each leaf node represents a class label. It is interpretable and capable of handling both numerical and categorical data. However, it is prone to overfitting, especially with small datasets.

A Decision Tree recursively splits the feature space based on feature values to minimize an impurity measure such as Gini Index or Entropy (Imran et al., 2025).

- Gini Index for node t :

$$Gini(t) = 1 - \sum_{i=1}^C p_i^2 \quad (1)$$

where p_i is the proportion of class i instances in node t .

- Entropy for node t :

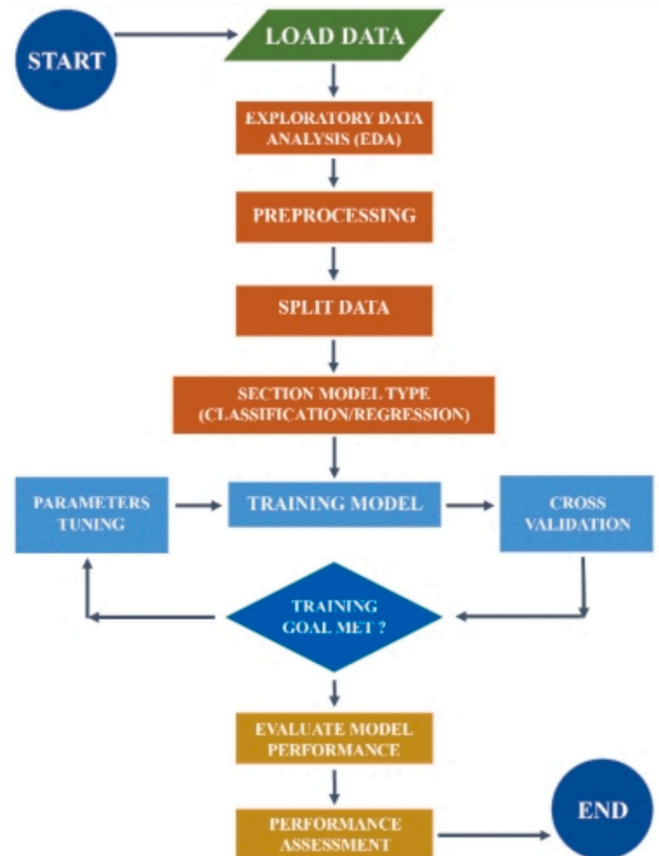


Fig. 1. The flow chart of general machine learning modeling.

$$Entropy(t) = - \sum_{i=1}^C P_i \log_2 P_i \quad (2)$$

- The split chosen at each node is the one that minimizes the weighted impurity after the split:

$$\Delta I = I(\text{parent}) - \sum_{k \in \text{children}} \frac{N_k}{N_{\text{parent}}} I(k) \quad (3)$$

where I is the impurity measure.

3.1.2. Random Forest (RF)

The Random Forest Classifier is an ensemble model that builds multiple decision trees during training and outputs the majority vote of the individual trees as the final prediction. It reduces overfitting and improves generalization by averaging multiple trees trained on random subsets of data and features.

3.1.3. XGBoost

XGBoost is a powerful boosting algorithm that sequentially builds trees, where each new tree attempts to correct the errors of the previous ones. It uses a gradient descent optimization framework and includes regularization to prevent overfitting, making it highly effective for structured tabular data.

The XGBoost technique is derived as follows

- At iteration t , prediction is updated as:

$$\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + f_t(x_i), \quad f_t \in F \quad (4)$$

where, f_t is the newly added tree and $F = \{f(x) = wq(x) \mid q : \mathbb{R}^m \rightarrow \mathbb{T}, w \in \mathbb{R}\}$ indicates a function space that defines a decision tree, and $\mathbb{T}$ is the leaf nodes of a decision tree. The following is the loss function:

$$L^{(t)} = \sum_{i=1}^n l(y_i, \hat{y}_i^{(t)}) + \sum_{k=1}^t \Omega(f_k) \quad (5)$$

$$\Omega(f) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2 \quad (6)$$

The first component in Eq. (5) presents the number of leaves, while the second component is the size of the outcome. XGBoost calculates Gain for every node in the tree to assess whether the generated branch is relevant.

$$\text{Gain} = \frac{1}{2} \left[\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} + \frac{(G_L + G_R)^2}{H_L + H_R + \lambda} \right] - \gamma \quad (7)$$

where $\frac{(G_L + G_R)^2}{H_L + H_R + \lambda}$ denotes the authentic gain before splitting and $-\gamma$ is the number of the new leaves.

3.1.4. LightGBM (Light Gradient Boosting Machine)

LightGBM is another gradient boosting framework that builds trees using a leaf-wise growth strategy with histogram-based decision rules, significantly improving training speed and memory efficiency. It is particularly suitable for large datasets and offers competitive accuracy.

LightGBM utilizes two combined techniques: Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB), which address the limitations of the histogram-based algorithm used in all GBDT (Gradient Boosting Decision Tree) frameworks (Marciano-Cedeno et al., 2010). In GOSS, different data instances contribute differently to calculating information gain, with instances having higher gradients contributing more. GOSS retains instances with high gradients while randomly dropping those with lower gradients to preserve the accuracy

of information gain estimation (Osman et al., 2021). The mathematical analysis of GOSS is presented in Eq. (8).

$$\hat{V}_j(d) = \frac{1}{n} \left(\frac{\left(\sum_{x_i \in A_l} g_i + \frac{1-a}{b} \sum_{x_i \in B_l} g_i \right)^2}{n_l^l(d)} + \frac{\left(\sum_{x_i \in A_r} g_i + \frac{1-a}{b} \sum_{x_i \in B_r} g_i \right)^2}{n_r^l(d)} \right) \quad (8)$$

where $\hat{V}_j(d)$ estimated variance gain over the subset $\cup B$, $A_l = \{x_i \in A: x_{ij} \leq d\}$, $A_r = \{x_i \in A: x_{ij} > d\}$, $B_l = \{x_i \in B: x_{ij} \leq d\}$, $B_r = \{x_i \in B: x_{ij} > d\}$, and the coefficient $1 - ab$ is used to normalize the sum of the gradients over B back to the size of A_c . Thus, the estimated $\hat{V}_j(d)$ issued over a smaller instance subset, instead of the accurate $V_j(d)$ over all the instances, to determine the split point. At the same time, the EFB technique is used by LightGBM to minimize the model complexity by bundling the exclusive features into a single feature.

3.1.5. KNeighbors (K Nearest Neighbors)

The KNN algorithm classifies a sample based on the majority class among its k nearest neighbors in the feature space. It is simple and non-parametric, relying on distance metrics (e.g., Euclidean) for classification. However, it can be sensitive to irrelevant features and scaling.

The KNN function employs Euclidean distance, which can be calculated with the following equation (Zhang, 2016):

$$D(p, q) = \sqrt{(p_1 - q_1)^2 + (p_2 - q_2)^2 + \dots + (p_n - q_n)^2} \quad (9)$$

where p and q are subjects to be compared with n characteristics. There are also other methods to calculate distance, such as Manhattan distance.

3.1.6. Support Vector Machine (SVM)

SVM constructs hyperplanes in a high-dimensional space to separate classes with maximum margin. It is effective for both linear and non-linear classification using kernel functions. SVM is robust in high-dimensional spaces and performs well when the number of features exceeds the number of samples.

The QP formulation for SVM classification. This is a simple representation only (Jakkula, 2006).

$$\min_{f, \xi} \|f\|_K^2 + C \sum_{i=1}^l \xi_i \quad (10)$$

Subject to

$$y_i f(x_i) \geq 1 - \xi_i \text{ for all } i \quad \xi_i \geq 0$$

SVM classification, Dual formulation:

$$\min_{\alpha_i} \sum_{i=1}^l \alpha_i - \frac{1}{2} \sum_{i=1}^l \sum_{j=1}^l \alpha_i \alpha_j y_i y_j K(x_i, x_j) \quad (11)$$

Subject to

$$0 \leq \alpha_i \leq C, \text{ for all } i; \quad \sum_{i=1}^l \alpha_i y_i = 0$$

Variables ξ_i are called slack variables and they measure the error made at point (x_i, y_i) .

Training SVM becomes quite challenging when the number of training points is large. Several methods for fast SVM training have been proposed.

3.2. Adaptive Boosting (Adaboost) model

AdaBoost combines multiple weak learners (typically decision stumps) into a strong classifier by adjusting the weights of misclassified instances in each iteration. It focuses on difficult examples, improving overall model accuracy while maintaining simplicity and interpretability.

Input required: TDS: Training Dataset $TDS = x_i(1, 2, 3, 4, \dots, n)$ labels $y_i \in Y$.

It: Iteration Number

Steps a to h

(a) Assign TDS sample $(x_i, y_i), \dots, (x_n, y_n); x_i \in X, y_i \in \{-1, +1\}$

(b) Initialize weights of $D_{it}(i) = 1/M, i = 1, \dots, M$

(c) For $It = 1, \dots, T$

(d) Train the learner that is weak using distribution D_{it}

(e) Get hypothesis of weak $h_{it} : X \rightarrow \{-1, +1\}$ along with its error =

$$\sum_{h_{it}(x) \neq y_i} D_{it}(i)$$

(f) Update distribution $D_{it} : D_{it+1}(i) = D_{it}(i) \exp(-\alpha_{it} y_{it} T_{it}(x_{it})) / C_{it}$

(g) Next It that, $It + 1$

(h) Final hypothesis Outputs:

$$H(x) = \text{sign} \left[\sum_{It=1}^T \alpha_{it} h_{it}(x) \right] \quad (12)$$

And,

$$\alpha_{it} = \frac{1}{2} \ln \left(\frac{P_{+1} - P_{-1}}{P_{-1} + P_{+1}} \right) \quad (13)$$

C_{it} is the normalization constant; α_{it} is used to allow the outcome to be generalized and to solve the problem of overfitting and noise-sensitive situations. The real value of $\alpha_{it} h_{it}(x)$ is built using a class probability estimate (P) (Mahesh et al., 2022)

3.3. Regression models for predicting WQI

In addition to classification, regression models were used to predict the numerical WQI values directly. The models were trained on the same set of input features and evaluated using performance metrics such as Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R-squared (R^2).

3.3.1. Linear regression

Linear regression models the relationship between the input variables and the WQI as a linear combination. It serves as a baseline model, assuming a linear dependency between predictors and the target. Its simplicity and interpretability make it useful, though it may underperform on non-linear relationships. Simple Linear regression defines the dependence of the variable. Simple regression distinguishes the influence of independent variables from the interaction of dependent variables (Maulud and Abdulazeez, 2020)

$$y = \beta_0 + \beta_1 x + \varepsilon \quad (14)$$

Multivariate linear regression (MLR): The basic model for MLR is:

$$y = \beta_0 + \beta_1 x_1 + \dots + \beta_m x_m + \varepsilon \quad (15)$$

The formula to determine the formula matrix is:

$$\hat{\beta} = (X^T X)^{-1} X^T y \quad (16)$$

$$\text{where, } \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \dots \\ \beta_m \end{bmatrix}, X = \begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1m} \\ 1 & x_{21} & x_{22} & \dots & x_{2m} \\ \dots & \dots & \dots & \dots & \dots \\ 1 & x_{m1} & x_{m2} & \dots & x_{nm} \end{bmatrix}, Y = \begin{bmatrix} Y_1 \\ Y_2 \\ \dots \\ Y_n \end{bmatrix}$$

Polynomial Regression: The Model of polynomial is

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \dots + \beta_h x^h + \varepsilon \quad (17)$$

where h is named the polynomial degree

3.3.2. Decision tree regressor

The Decision Tree Regressor follows a tree-based structure to predict continuous WQI values by partitioning the feature space into distinct regions. Each region corresponds to a mean value of the target within that subset. It can model non-linear relationships but may be overfit without pruning.

In regression problems, the goal is to find a tree structure that minimizes the mean squared (or absolute) error of the predictions (Yazdani et al., 2022) as defined in Eq. (18)

$$MSE = \frac{1}{d \times k} \sum_{y \in \bar{y}} (y - \bar{y})^2 \quad (18)$$

At node n of a decision tree, a dataset $D^n = (X^n, y^n)$ with N_n number of samples is available. For these samples, $\theta_n = (f, t_n)$ splits the data based on the threshold t_n in feature f into left ($-$) and right ($+$) categories as in Eqs. (19) and (20), respectively. To determine the optimal θ_n , a function as defined in Eq. (21) is utilized that gives a quantitative evaluation of each split. In this metric, function H is determined based on the type of problem (regression or classification). In the case of our regression problem, the mean squared error as defined in Eq. (22) can assess the quality of each cut. Finally, minimizing over θ_n , the optimal cut is achieved. This process is repeated to grow the tree until no improvement in the loss function as in Eq. (18) is reached

$$D^{n-}(\theta_n) = \left\{ (X^n, y^n) \mid X_f^n < t_n \right\} \quad (19)$$

$$D^{n+}(\theta_n) = D^n - D^{n-}(\theta_n) \quad (20)$$

$$G(D^n, \theta_n) = \frac{N_n^-}{N_n} H(D^{n-}(\theta_n)) + \frac{N_n^+}{N_n} H(D^{n+}(\theta_n)) \quad (21)$$

$$H(D^n) = \frac{1}{N_n} \sum_{y \in \bar{y}} (y - \bar{y})^2 \quad (22)$$

$$\theta^{opt} = \arg_{\theta_n} \min G(D^n, \theta_n) \quad (23)$$

3.3.3. Random Forest Regressor

The Random Forest Regressor uses an ensemble of decision trees and outputs the average of individual predictions. It improves accuracy and reduces overfitting, especially in the presence of noise and complex interactions between features.

The general prediction method of the RFR model is (Cheng et al., 2023):

- (1) Randomly take samples from training samples ($n \times \text{sample}$) for n times to form a training set (samples were put back after every sampling). Repeat r times to obtain training sets: $D_1, D_2, \dots, D_r$.
- (2) For each training set, k attributes are randomly selected from the attribute set ($m \times \text{attribute}$), $k = \log(2m)$, and then cart trees are established: $f_1(x), f_2(x), \dots, f_r(x)$.
- (3) The final prediction value of the random forest is determined by the average method:

$$f(x) = \frac{1}{r} \sum_{i=1}^r f_i(x) \quad (24)$$

3.3.4. XGBoost Regressor

XGBoost for regression follows the same boosting principle as in classification but optimizes for mean squared error in predicting continuous WQI values. It is efficient, scalable, and suitable for modeling complex, non-linear patterns in the data.

The XGBoost regression tree model is expressed mathematically as (Zou et al., 2022)

$$\hat{y}_i = \alpha \sum_{k=1}^K f_k(x_i) \quad (25)$$

where $\hat{y}_i$ is the predicted value corresponding to the input x_i , α is the learning rate of the individual regression tree, K is the total number of CARTs (classification and regression trees) being used, and f_k is the output of the k -th regression tree. Eq. (25) shows that the predicted score $\hat{y}_i$ is the sum of all f_k values.

After obtaining the prediction result, the objective function L was used to evaluate the quality of the obtained results, denoted as

$$L = \sum_i^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k) \quad (26)$$

3.3.5. KNN Regression Model

KNN regression predicts the WQI of a sample by averaging the target values of its k nearest neighbors. It does not make strong assumptions about the functional form of the data but is sensitive to outliers and computationally intensive with large datasets.

The KNN calculation uses the average of the numerical target of the K nearest neighbors, applying one of the following distance functions (Elmezughi et al., 2022):

$$\text{Euclidean} : D = \sqrt{\sum_{i=1}^K (X_i - Y_i)^2} \quad (27)$$

$$\text{Manhattan} : D = \sum_{i=1}^K |X_i - Y_i| \quad (28)$$

$$\text{Minkowski} : D = \left(\sqrt[q]{\sum_{i=1}^K (|X_i - Y_i|)^q} \right)^{1/q}, \quad q \geq 1 \quad (29)$$

where X and Y are the original path loss and the predicted path loss, respectively.

3.3.6. SVR Model

SVR extends SVM for regression tasks by fitting a function within a specified margin (epsilon-insensitive tube) around the true target values. It aims to minimize the model complexity while penalizing deviations beyond the epsilon threshold, making it effective for high-dimensional data.

Generally, the estimation function in SVR takes the following form

$$f(x) = (\omega \cdot \phi(x)) + b \quad (30)$$

In Eq. (31), $(\cdot)$ denotes the inner product in Ω , a feature space of possibly different dimensionality such that $\omega: X \rightarrow \Omega$ and $b \in \mathbb{R}$. The other two parameters, ω and b , can be determined from the training dataset by minimizing the regression risk based on the estimated risk (Mahabub and Habib, 2019)

3.3.7. MLP Regressor

The MLP Regressor is a type of feedforward artificial neural network that uses backpropagation for training. It can capture complex non-linear relationships between input variables and WQI. The network architecture includes input, hidden, and output layers with non-linear activation functions.

The computational scheme of the single neuron j :

$$x_j = \phi \left(\sum_i w_{ij} x_i + b_j \right) \quad \text{with, e.g., } \phi_j(u) \begin{cases} u & \text{if } u > 0 \\ 0.001u & \text{otherwise,} \end{cases} \quad (31)$$

where x_i refers to the input value or output of a neuron of the previous layer. w_{ij} refers to a trainable weight from neuron or input i to neuron j , b_j is a trainable bias value, and ϕ_j is the activation function, which computes the output of neuron j . In the overall network, an input pattern is propagated stage-wise from the input layer to the output layer.

4. Methodology proceeding

The dataset was preprocessed to ensure consistency before modeling. Missing values were imputed using the mean for normally distributed and the median for skewed parameters, thereby preserving data distribution. All features were then normalized using z-score standardization to eliminate scale differences and prevent dominance of high-magnitude variables. Feature selection was performed through Pearson correlation analysis, where variables with strong pairwise correlations ($|r| \geq 0.85$) were examined and redundant predictors removed. This approach reduced multicollinearity while retaining the most informative variables for model development (Fig. 2).

Multiple machine learning regressors were developed and compared in this study, including SVR and MLP neural networks. The structure of the SVR model, which depicts the regression process where the algorithm tries to fit the best function within a threshold margin. Meanwhile, the MLP neural network topology consists of an input layer, one or more hidden layers with activation functions, and an output layer. These models were trained using optimized parameters obtained via Grid Search with 5-fold cross-validation, which systematically explored different combinations of hyperparameters to identify the optimal configuration.

In this study, multiple machine learning models were employed for predictive analysis, while deep learning architectures like Convolutional Neural Networks (CNN) or LSTM networks were deliberately not considered. The primary reason is that the dataset consists of a relatively small number of samples (170 observations), which is insufficient to train deep neural networks effectively. Deep learning models typically require very large datasets to capture complex patterns without overfitting, whereas traditional machine learning methods perform well on small to medium-sized tabular datasets. Moreover, the input features in this study are structured physicochemical parameters rather than unstructured data such as images or time series, where CNNs or LSTMs generally excel. Machine learning models also offer greater interpretability through tools such as feature importance analysis and SHAP values, making them more suitable for hydrochemical studies where understanding variable influence is as important as prediction accuracy. Therefore, the choice of tree-based and ensemble machine learning algorithms provided a balance of robustness, efficiency, and interpretability, making them better suited than deep learning approaches for this dataset.

The complete flow of the study's approach, from data collection to model evaluation, is depicted in Fig. 2, which shows the proposed methodology. This includes data acquisition, preprocessing, WQI computation, model training and testing, and performance evaluation based on metrics like MAE, RMSE, and R^2 .

4.1. WQI computation

WQI is a dominant indicator that impacts the water quality (Tyagi et al., 2020). WQI is computed by utilizing various parameters. WQI is computed using. (32):

$$WQI = \frac{\sum_{i=1}^N q_i \times w_i}{\sum_{i=1}^N w_i} \quad (32)$$

where N represents the number of parameters, q_i represents the quality rating scale for parameter i , and w_i represents the unit weight for the parameter i . q_i is computed using Eq. (33):

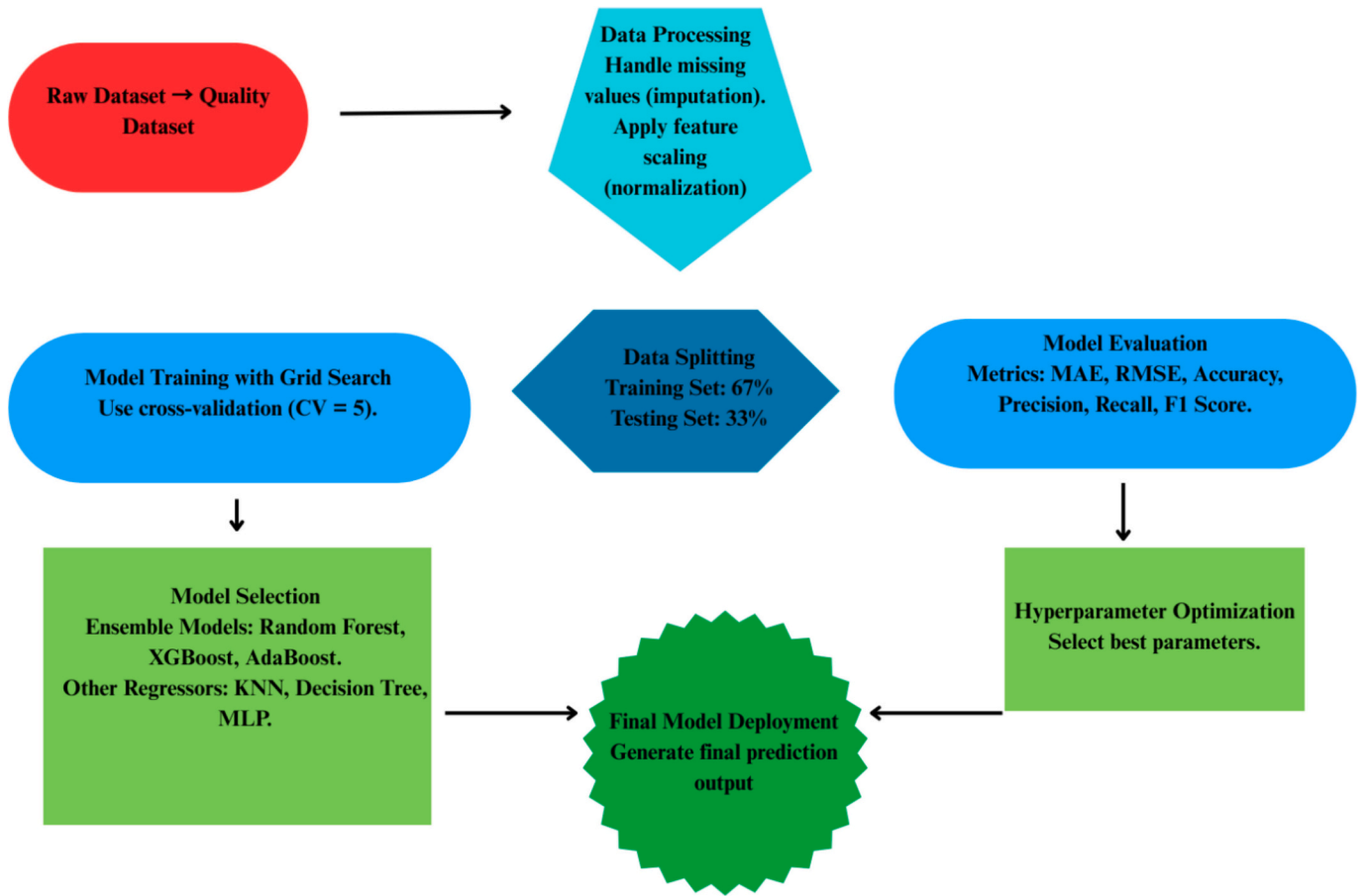


Fig. 2. The proposed methodology.

$$q_i = 100 \times \left(\frac{v_i - v_{id}}{s_i - v_{id}} \right) \quad (33)$$

where v_i represents the estimated value for the parameter i , v_{id} represents an ideal value for the parameter i while the water is pure, and s_i represents a standard value for the parameter i . The unit weight w_i is computed using Eq. (34):

$$w_i = \frac{k}{s_i} \quad (34)$$

Where k represents the constant of proportionality and is computed using Eq. (35):

$$k = \frac{1}{\sum_{i=1}^N s_i} \quad (35)$$

4.2. Operational early-warning protocol (real-time deployment)

Incoming sensor readings (DO (mg/L), BOD (mg/L), CO₂, pH, alkalinity, hardness, calcium, ammonia, nitrite, phosphorus, H₂S, plankton, turbidity, Temp) are standardized by the training-only pre-processor embedded in the persisted pipeline and scored by the LightGBM classifier. We compute the probability of poor water (class 2), $P(\text{poor})$ and apply a four-level decision scheme with predefined interventions. Rules: Normal if $P(\text{poor}) < 0.50$ and $\text{DO} \geq 5$ mg/l; Caution if $0.50 \leq P(\text{poor}) < 0.80$ or $5 > \text{DO} \geq 3$ mg/l; Warning if $P(\text{poor}) \geq 0.80$ or $\text{WQI} < 60$ or $\text{DO drop} > 1$ mg/l in 60 min; Critical if $\text{DO} < 3$ mg/l or $[P(\text{poor}) \geq 0.90] \wedge [\text{WQI} < 60]$. The pipeline is stateless (no refitting), preventing data leakage and enabling continuous real-time scoring.

On CPU (Intel® Xeon® 2.20 GHz, 13.6 GB RAM; Python 3.12;

NumPy 2.0; Pandas 2.2; scikit-learn 1.6; LightGBM 4.6), the end-to-end preprocess + predict latency of the persisted pipeline was milliseconds per sample and well under 1 s for typical batch sizes. Measured mean $\pm$ SD (5 runs per size) (Table 8).

4.3. Computational environment and implementation details

The experiments were carried out on Google Colab's free-tier platform using a CPU-only runtime environment running Linux kernel version 6.1.123+. The virtual machine was equipped with an Intel(R) Xeon(R) CPU operating at 2.20 GHz and 13.61 GB of RAM. The software stack included Python 3.12.11 along with key libraries such as NumPy 2.0.2, Pandas 2.2.2, Matplotlib 3.10.0, scikit-learn 1.6.1, XGBoost 3.0.5, and LightGBM 4.6.0. Seven machine learning models were implemented using these libraries: DecisionTreeClassifier, RandomForestClassifier, SVC, KNeighborsClassifier, and AdaBoostClassifier from scikit-learn; XGBClassifier from XGBoost; and LGBMClassifier from LightGBM. To ensure reproducibility, all random seeds were fixed using `random_state = 42`. The dataset was split using a stratified 67:33 train-test ratio, and model performance was further validated through stratified 5-fold cross-validation.

5. Results

The analysis reveals key insights into the relationships between water quality parameters and potability. The correlation heatmap shows that DO ($r = 0.31$), BOD ($r = 0.41$), and Phosphorus ($r = 0.39$) have a positive association with potability, while Turbidity ($r = -0.27$) shows a negative correlation, highlighting its strong adverse effect on water drinkability. Significant interrelations were also observed among

variables like BOD and Nitrite ($r = 0.47$) and Ammonia and Phosphorus ($r = 0.32$), indicating pollution-related nutrient interactions. Fig. 3 presents a heat map illustrating the correlations among features, highlighting the strength and direction of their relationships. Table 3 displays the statistical calculations of the features, including measures mean, standard deviation, minimum, and maximum values.

Outliers identified in parameters such as temperature, calcium, alkalinity, and plankton were retained to reflect natural spatial variability, with only erroneous values corrected or excluded, while normalization and robust algorithms minimized their impact (Fig. 4). High dispersion in Ammonia, Nitrite, and H₂S suggests local contamination spikes, which may influence potability ratings. Density plots across potability classes show clear patterns: Excellent quality water is associated with low BOD, CO₂, H₂S, Ammonia, and Nitrite, while high DO and balanced pH (7–8) are consistent markers of better potability (Fig. 5). In contrast, poor potability corresponds with higher concentrations of pollutants and reduced oxygen levels.

SHAP-based feature importance with class-specific beeswarm plots and a global ranking (mean |SHAP|) (Fig. 6). Across methods H₂S, Alkalinity, DO, CO₂, BOD, Turbidity, Ammonia, and Nitrite consistently

rank among the top contributors.

Re-estimating pairwise associations on the original data, DO and BOD exhibited a positive relationship: Pearson $r = 0.292$, Spearman $\rho = 0.476$ ($p = 5.22 \times 10^{-242}$), and partial $r \approx 0.113$ after controlling for the other measured parameters (Fig. 7). Full tables and robustness checks are provided in the Supplement. Each point is a sample; color encodes BOD (mg/L). Solid lines show a linear fit (black) and LOWESS smooth (cyan). Annotated statistics: Pearson $r = 0.292$, Spearman $\rho = 0.476$ ($p = 5.22 \times 10^{-242}$), partial $r \approx 0.113$ (controlling for all other measured numeric variables). Although an inverse DO–BOD relation is typical in rivers, the positive association here is consistent with managed aquaculture/pond settings where aeration, daytime sampling, and water exchange maintain elevated DO under higher organic load.

5.1. WQC prediction

The classification models' performance was assessed using assessment criteria such as accuracy, recall, precision, F1 score, and Matthew's correlation coefficient (MCC). Eq. (36) is used to calculate precision:

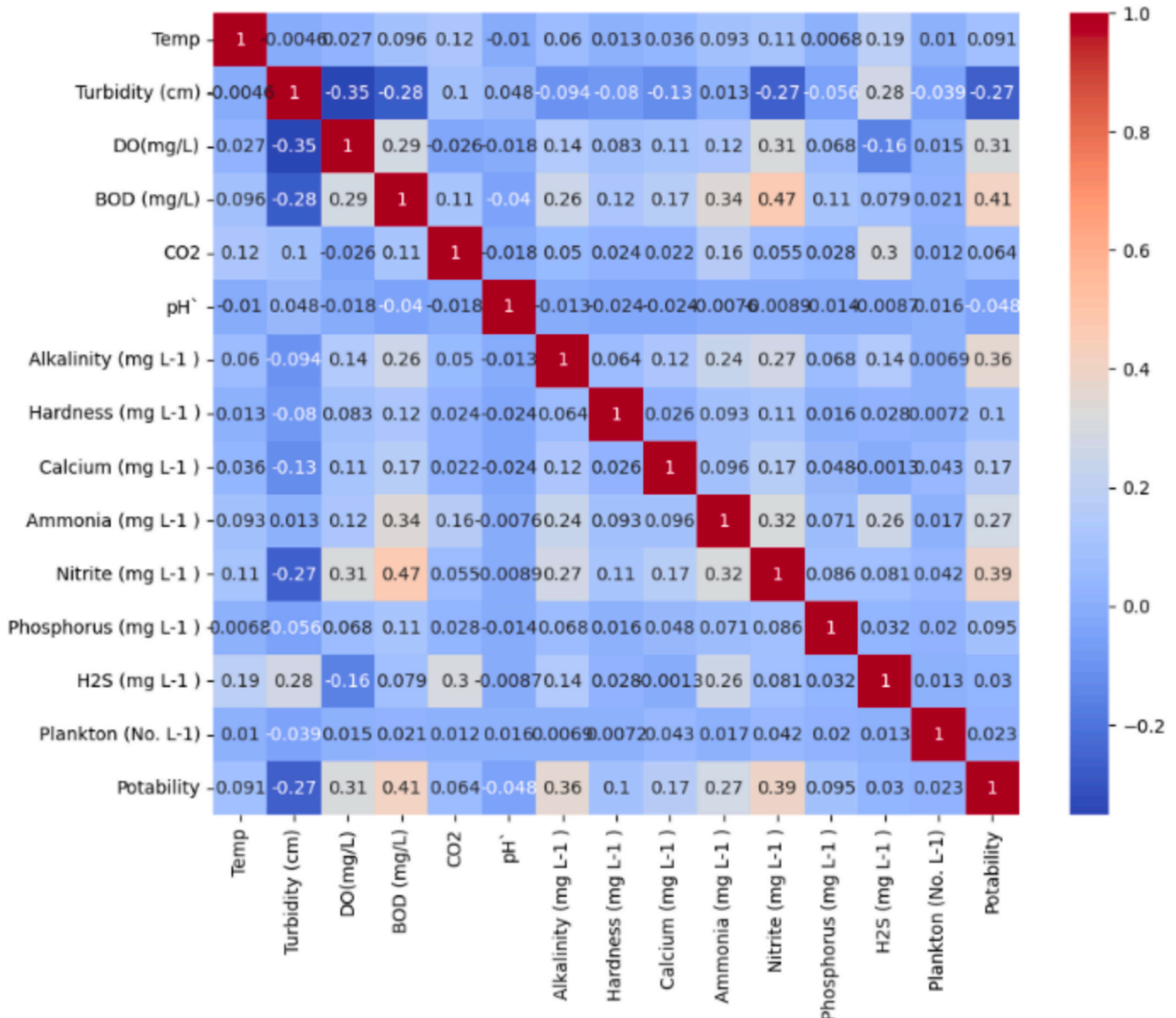
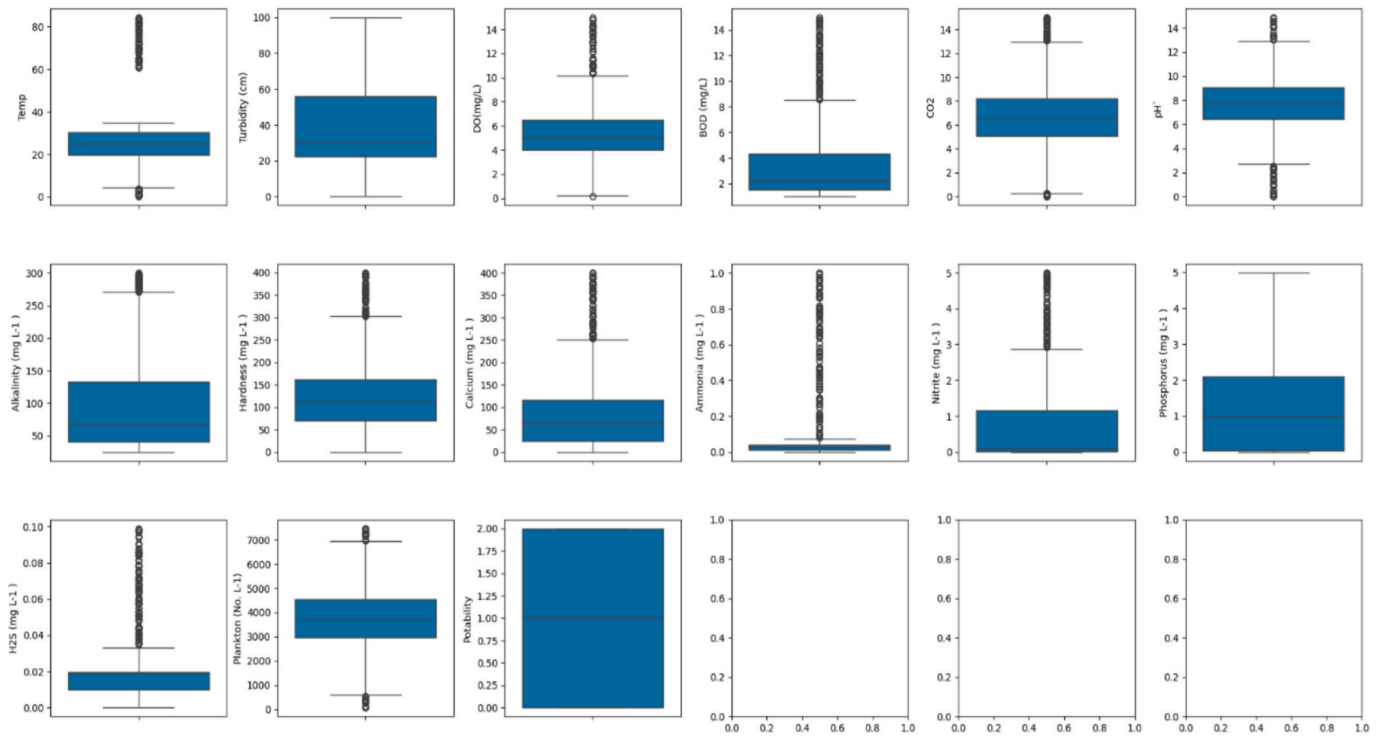


Fig. 3. Heat map visualization of the feature correlations.

Table 3

Statistical calculation of the features.

Parameters	Count	Mean	Std	Min	25 %	50 %	75 %	Max
Temperature	4300.0000	25.695695	9.670181	0.190000	19.777500	25.045000	30.275000	84.250000
Turbidity(cm)	4300.0000	39.046642	20.942694	0.050000	22.220000	30.210000	55.942500	99.800000
DO (mg/L)	4300.0000	5.300355	1.832741	0.134000	3.980000	5.001000	6.521050	14.970000
BOD (mg/L)	4300.0000	3.125079	2.292058	1.000000	1.520000	2.236000	4.324600	14.943000
CO ₂	4300.0000	6.375921	2.831105	0.001000	5.050000	6.600000	8.241625	14.984000
pH	4300.0000	7.713461	1.580273	0.004000	6.443000	7.740000	9.035175	14.851000
Alkalinity (mg/L)	4300.0000	93.717099	68.949092	25.01200	40.422300	67.565000	132.834100	299.91300
Hardness (mg/L)	4300.0000	127.055445	78.883090	0.256000	69.479275	111.06000	162.675625	398.797000
Calcium (mg/L)	4300.0000	84.871948	75.718836	0.018000	23.745100	62.843000	115.596850	399.321000
Ammonia (mg/L)	4300.0000	0.048268	0.122891	0.000000	0.013000	0.026300	0.039000	0.999000
Nitrate (mg/L)	4300.0000	0.642947	0.903594	0.000000	0.011000	0.099900	1.166750	4.990000
Phosphorus (mg/L)	4300.0000	1.172745	1.082642	0.000000	0.028275	0.975000	2.101325	4.974000
H ₂ S (mg/L)	4300.0000	0.016472	0.011872	0.000000	0.010000	0.019000	0.019600	0.099000
Plankton (No. L ⁻¹)	4300.0000	3805.508432	1208.537953	79.00000	2956.0000	3729.5000	4555.0000	7460.0000
Potability	4300.0000	1.023256	0.820996	0.000000	0.000000	1.000000	2.000000	2.000000

Note: Std- Standard deviation, Min- Minimum, Max- Maximum.**Fig. 4.** Box plots for all the water quality parameters.

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + FN + TN} \quad (36)$$

where TP is true positive, TN is true negative, FP is false positive, and FN is false negative. Recall is calculated using Eq. (37):

$$\text{Recall} = \frac{TP}{TP + FN} \quad (37)$$

Precision is calculated using Eq. (38):

$$\text{Precision} = \frac{TP}{TP + FP} \quad (38)$$

F1 score is computed using Eq. (39):

$$\text{F1Score} = \frac{2 * \text{Recall} * \text{Precision}}{\text{Recall} + \text{Precision}} \quad (39)$$

MCC is calculated using Eq. (40):

$$\text{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (40)$$

MAE, MedAE, MSE, and R² were used to assess the effectiveness of the regression models. Eq. (41) is used to compute MAE:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_{\text{real}_i} - y_{\text{pred}_i}| \quad (41)$$

MedAE is calculated using Eq. (42):

$$\text{MedAE} = \text{median}(|y_{\text{real}_1} - y_{\text{pred}_1}|, \dots, |y_{\text{real}_N} - y_{\text{pred}_N}|) \quad (42)$$

MSE is calculated using Eq. (43):

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (y_{\text{real}_i} - y_{\text{pred}_i})^2 \quad (43)$$

R2 is calculated using Eq. (44):

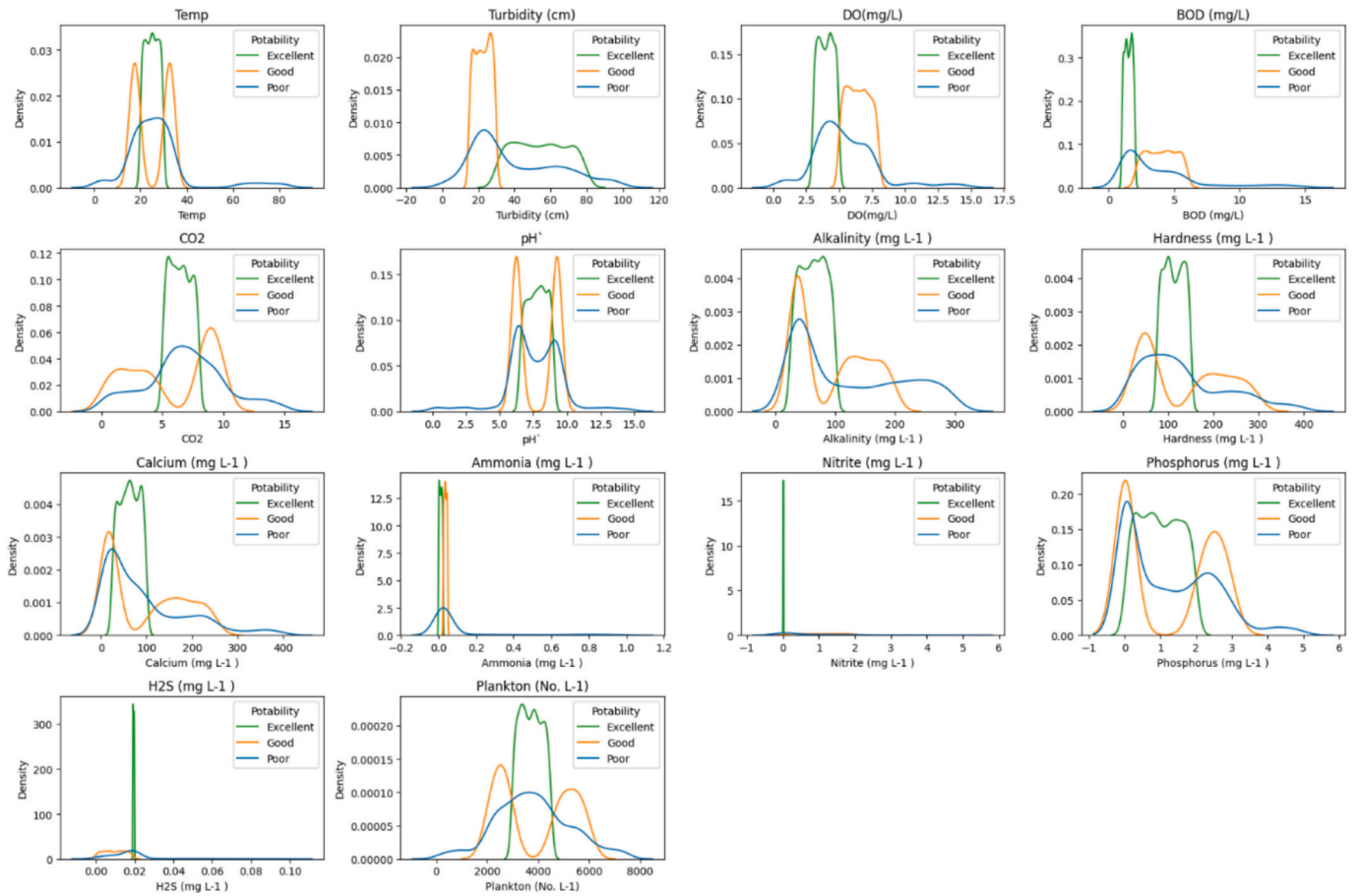


Fig. 5. Density plot for water quality parameter with potability.

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_{real_i} - y_{pred_i})^2}{\sum_{i=1}^N (y_{real_i} - \bar{y})^2} \quad (44)$$

In evaluating predictive models for water quality, performance metrics such as the R^2 , accuracy, and the F1-score hold practical importance beyond statistical interpretation. The R^2 value reflects how well the model explains the variability of water quality parameters. A high R^2 indicates that the model can reliably capture hydrochemical dynamics, which is essential for decision-makers to anticipate changes in water hardness, nutrient loads, or pollution risks with confidence. Accuracy, which measures the proportion of correctly classified outcomes, is particularly valuable when models are used for categorical assessments such as determining whether water meets drinking or irrigation standards. High accuracy ensures that monitoring systems correctly identify safe versus unsafe conditions, reducing the risk of public health hazards or economic losses in aquaculture and agriculture. The F1-score, as the harmonic mean of precision and recall, becomes especially critical in imbalanced datasets where certain water quality conditions (e.g., harmful algal bloom events or extreme hardness levels) are rare but carry serious consequences. A strong F1-score demonstrates that the model not only avoids false alarms but also minimizes the chance of missing critical events. Collectively, these metrics ensure that predictive models provide reliable, actionable insights, supporting proactive water management policies and targeted interventions.

The candidate ranges for hyperparameter tuning were defined based on commonly reported default values in the scikit-learn, XGBoost, and LightGBM documentation, as well as insights from prior studies applying similar models to water quality datasets. For instance, ranges such as $n_{estimators} = \{100, 200\}$ and $max_depth = \{4, 6\}$ were selected to balance coverage with computational feasibility, given the modest sample

size and the need to evaluate multiple models simultaneously. Accuracy was used as the primary optimization metric during tuning because the dataset is relatively balanced and accuracy provides a straightforward global measure of classification performance. To ensure that reliance on accuracy did not obscure potential class-specific trade-offs, additional metrics such as precision, recall, F1-score, and Matthews Correlation Coefficient (MCC) were also reported on the held-out test set. These complementary measures confirmed that models optimized on accuracy maintained strong and consistent performance across multiple evaluation dimensions. Although exhaustive grid search was suitable for this study, we recognize its limitations in larger or higher-dimensional parameter spaces. The restricted grids employed here were designed to remain computationally feasible while capturing the most influential hyperparameters.

Table 4 outlines the optimal parameter settings for the classification approaches, determined using the grid search algorithm. The performance comparison of classification algorithms revealed that LightGBM achieved the highest accuracy (0.9965), precision (0.9965), recall (0.9965), F1 score (0.9965), and MCC (0.9947), making it the most effective model in this study. XGBoost and Random Forest also demonstrated excellent results, with XGBoost slightly outperforming Random Forest across most metrics (accuracy of 0.9930 and MCC of 0.9895 for XGBoost; accuracy of 0.9901 and MCC of 0.9852 for Random Forest) (Fig. 9). These results highlight the superior performance of boosting and ensemble-based approaches (Table 5). In contrast, the Decision Tree classifier, although relatively strong, showed lower performance (accuracy and other metrics at 0.9732; MCC at 0.9599), reflecting the benefits of ensemble learning present in its more advanced counterparts. SVM and KNN yielded moderate performance, with SVM performing better (accuracy of 0.9549 and MCC of 0.9341) than KNN

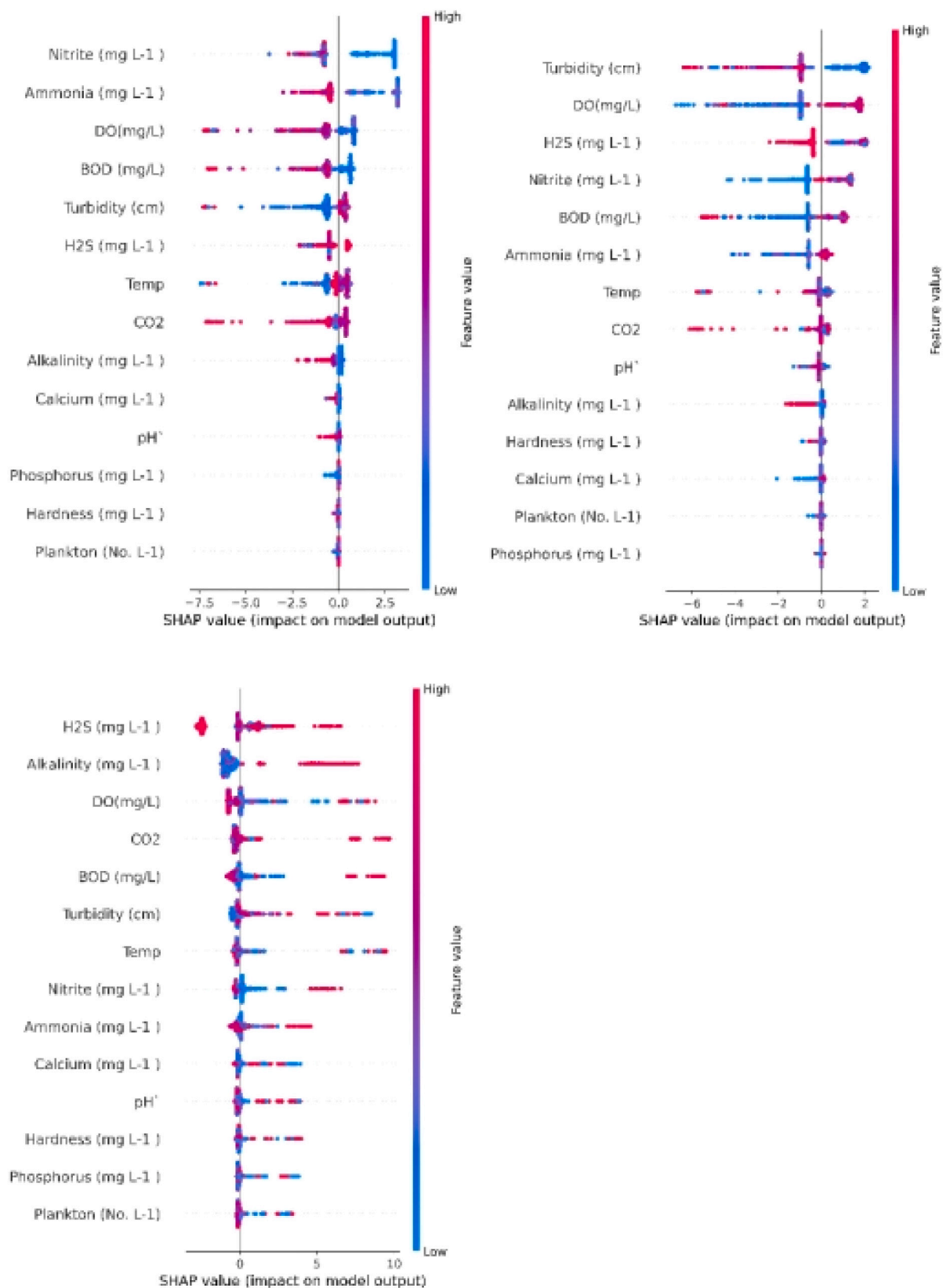


Fig. 6. Shape value of Water Portability Index (0,1,2).

(accuracy of 0.8626 and MCC of 0.8124), likely due to their sensitivity to data distribution and dimensionality. AdaBoost was the weakest performer among the models, with the lowest accuracy (0.7710), F1 score (0.7413), and MCC (0.6887), suggesting limited capability in

handling complex classification tasks in this context. Overall, the results suggest that modern gradient boosting methods, such as LightGBM and XGBoost, are highly suitable for high-precision classification tasks due to their strong predictive performance and generalization capabilities.

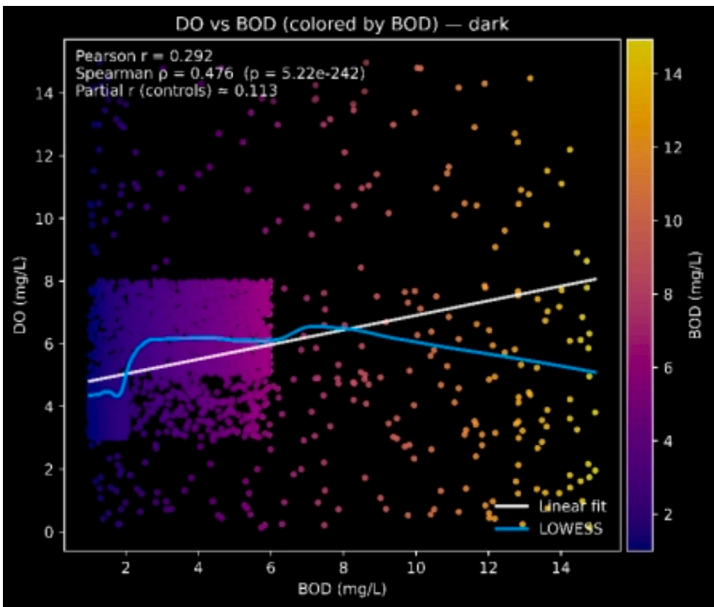


Fig. 7. DO–BOD association: scatter, linear fit, and LOWESS smooth.

Table 4
The settings of the best parameters for the classification approaches using the grid search algorithm.

Approaches	Parameters tuning	The best parameters
XGBoost	'n_estimators': [100,200], 'max_depth': [4, 6], 'learning_rate': [0.05, 0.1], 'subsample': [0.8,0.9], 'colsample_bytree': [0.8,0.9]	'n_estimators': 100, 'learning_rate': 0.1, 'max_depth': 4, 'subsample': 0.9, 'colsample_bytree': 0.8
SVM	C: [0.1, 1, 10]	C: 10
Decision Tree (DT)	kernel = ['linear', 'rbf', 'poly']	Kernel: 'rbf'
KNeighbor	'max_depth': [8, 12], 'min_samples_split': [2, 5] n_neighbors: [5, 7]	max_depth: 12, min_samples_split: 2 n_neighbors: 5
LightGBM	'num_leaves': [31, 50, 70], 'max_depth': [-1, 15, 20], 'learning_rate': [0.05, 0.1], 'n_estimators': [100, 200], 'min_child_samples': [1, 5, 10], 'subsample': [1.0,0.9], 'colsample_bytree': [1.0,0.9]	'num_leaves': 70, 'max_depth': -1, 'learning_rate': 0.1, 'n_estimators': 200, 'min_child_samples': 10, 'subsample': 1.0, 'colsample_bytree': 0.9,
RF	n_estimators: [100, 200], max_depth: [10, 15], min_samples_split: [2, 5]	n_estimators: 200 max_depth: 15, min_samples_split: 5,
AdaBoost	'n_estimators': [50,100, 150], 'learning_rate': [0.1,0.3, 0.5]	n_estimators: 50, learning_rate: 0.5

Comprehensive overfitting diagnostics were conducted to evaluate the robustness of the predictive models. Learning curves, generated by plotting training accuracy against five-fold stratified cross-validation (CV) accuracy with imputation and scaling embedded within the CV pipeline, showed consistent convergence. As the sample size increased, the CV curve approached the training curve with only a small and stable gap (train $\approx$ 0.999–1.000 vs. CV $\approx$ 0.991–0.994; $\Delta \approx$ 0.006–0.009),

indicating strong generalization without signs of overfitting (Fig. 8). To further validate model stability, Repeated Stratified 5 $\times$ 3-fold CV produced an average accuracy of 0.9928 ± 0.0040 , demonstrating resilience to different resampling splits. Performance on the independent test set was also consistent, with accuracy = 0.9923 and macro-F1 = 0.9924, corresponding to only 11 misclassifications out of 1419 samples. Class-level evaluation confirmed balanced predictive power, with precision/recall values of 0.9978/1.0000 for class 0, 0.9866/0.9910 for class 1, and 0.9923/0.9866 for class 2. These results confirm that the models achieved strong but not artificially perfect performance, reinforcing confidence in their generalization ability. Importantly, all procedures were designed to prevent data leakage, including strict stratified train/test splits, fitting of scalers on training folds only, and exclusion of the test set from model selection and early stopping. In water quality monitoring, R^2 indicates how well a model predicts continuous variables like pollutant levels—higher values mean better forecasting. Accuracy shows how often a classification model correctly labels water as “safe” or “unsafe,” but it can be misleading if the data is imbalanced. F1-score balances precision and recall, making it ideal for detecting rare but critical contamination events. Together, these metrics help ensure reliable predictions, timely alerts, and informed decisions for protecting public health and managing water resources effectively.

5.2. WQI prediction

Table 6 presents the best parameter configurations for the regression models, identified through the grid search method. In this section, the predictive performance of various regression models for WQI estimation was evaluated using optimal hyperparameters derived from grid search. The performance metrics—MAE, MSE, MedAE, and R-squared (R^2)—are presented in Table 7.

Table 5
The performance of the classification approaches using the grid search algorithms.

	Decision Tree	Random Forest	XGBoost	LightGBM	KNeighbors	SVM	AdaBoost
Accuracy	0.973	0.990	0.993	0.996	0.862	0.954	0.771
Precision	0.973	0.990	0.993	0.996	0.887	0.957	0.799
Recall	0.973	0.990	0.993	0.996	0.862	0.954	0.771
F1 Score	0.973	0.990	0.992	0.996	0.853	0.954	0.741
MCC	0.959	0.985	0.989	0.994	0.812	0.934	0.688

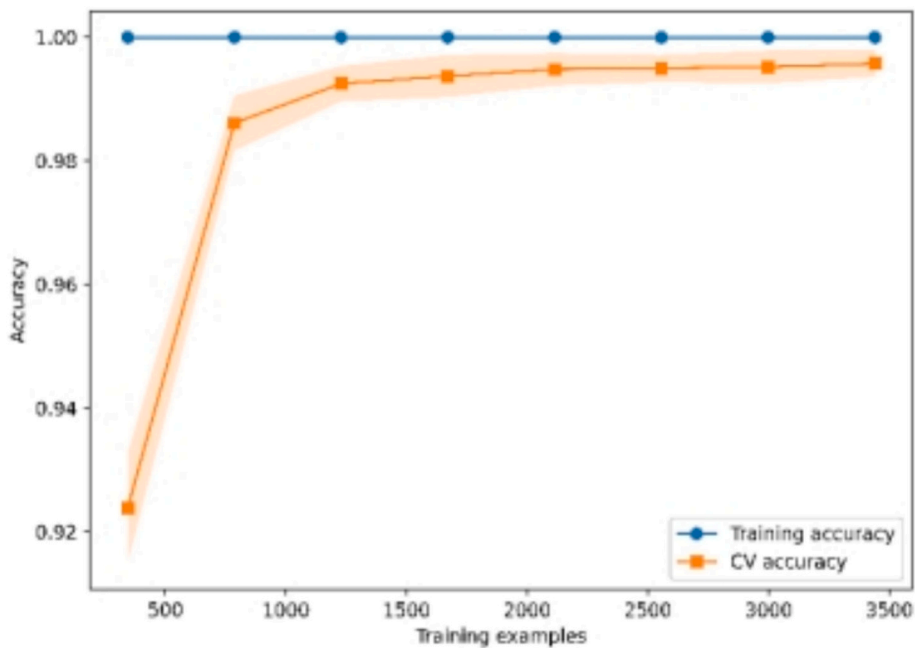


Fig. 8. Learning curves LightGBM.

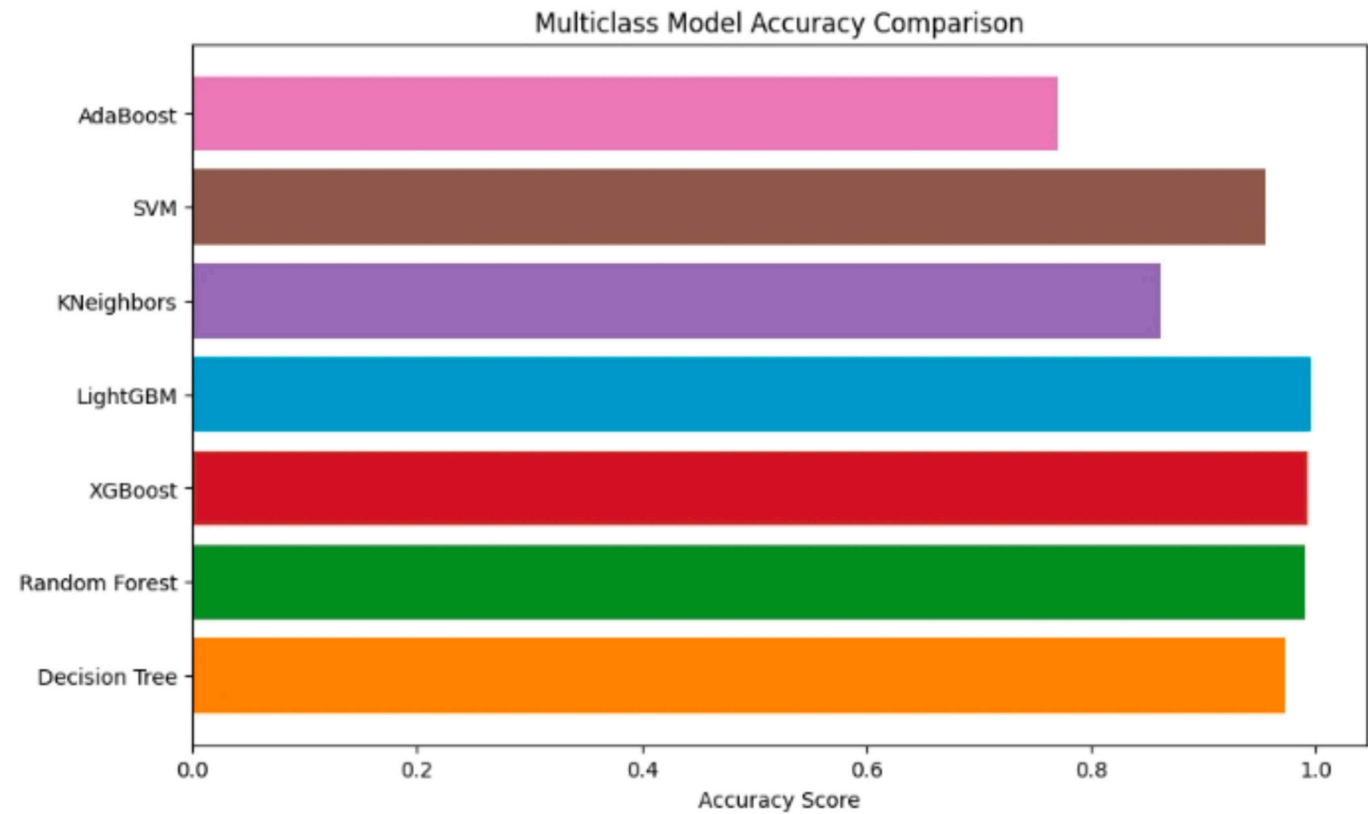


Fig. 9. Comparison between AdaBoost, SVM, KNeighbors, XGBoost model, Random Forest, Decision Tree, and LightGBM model in terms of accuracy.

Among all models, the XGBoost Regressor demonstrated the best overall performance, achieving the lowest MAE (0.0677), MSE (0.0215), and MedAE (0.0076), along with the highest R^2 value of 0.9685, indicating superior accuracy and generalization capability. This can be attributed to XGBoost’s ability to handle non-linearity and complex interactions in the dataset through boosting and regularization

techniques.

The Random Forest Regressor also performed exceptionally well, with an MAE of 0.0778, MSE of 0.0370, and R^2 of 0.9458. Its ensemble structure and ability to reduce overfitting through bootstrapping and averaging make it a strong candidate for WQI prediction tasks, though slightly less precise than XGBoost.

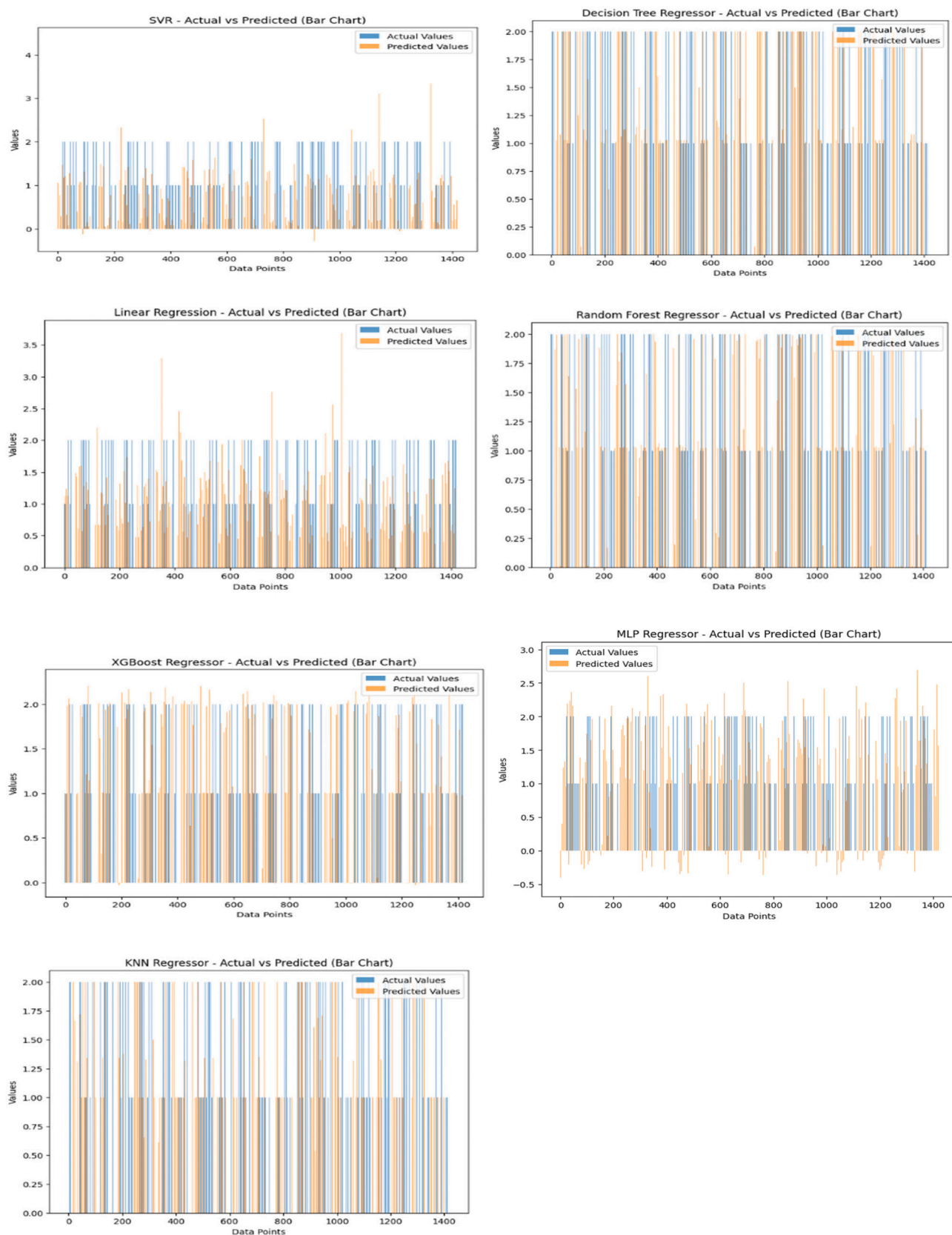


Fig. 10. Actual values vs predicted values for different Regressor models.

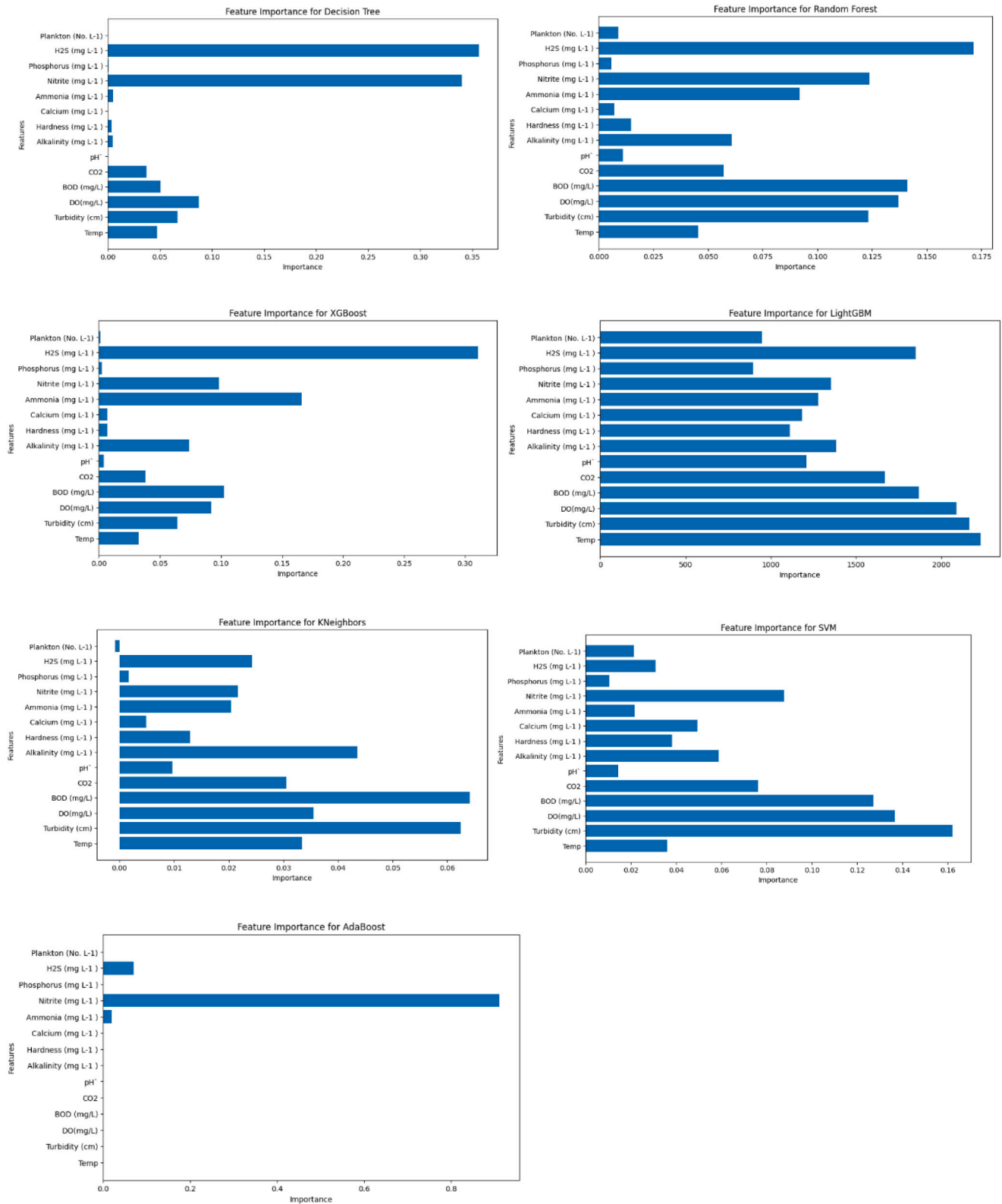


Fig. 11. Feature importance for ML model.

The Decision Tree Regressor produced decent results ($R^2 = 0.8437$), but its performance was notably less robust compared to the ensemble models. While it performed well on the training data, its lower R^2 value and higher MAE indicate potential overfitting or sensitivity to noise. The MLP Regressor, a neural network-based model, showed moderate performance with an R^2 of 0.7889. This suggests that while MLP can model

non-linear relationships, it might require deeper architectures or more training data for better accuracy in this specific context.

On the other hand, KNN Regressor and SVR yielded relatively poor performance. KNN had an MAE of 0.1784 and R^2 of 0.7442, which suggests it struggled with generalizing patterns in the data. SVR performed the worst among all models, with a high MAE (0.5415), MSE

Table 6
Best parameters for the regression models using the grid search method.

Approaches	Parameters tuning	The best parameters
Linear Regression	'fit_intercept': [True, False] 'max_depth': [5, 10, 15], 'min_samples_split': [2, 5, 10],	'fit_intercept': True
Decision Tree Regressor	'min_samples_leaf': [1, 2, 4], 'max_features': ['sqrt', 'log2'] n_estimators': [100, 200], 'max_depth': [10, 15], 'min_samples_split': [2, 5],	'max_depth': 15, 'max_features': 'sqrt', 'min_samples_leaf': 4, 'min_samples_split': 5
Random Forest Regressor	'min_samples_leaf': [1, 2], 'max_features': ['sqrt', None] 'bootstrap': [True, False] 'n_estimators': [100, 200], 'max_depth': [3, 6], 'learning_rate': [0.01, 0.1], 'subsample': [0.7, 0.8, 0.9, 1.0], 'colsample_bytree': [0.7, 0.8, 1.0] 'gamma': [0, 0.1, 0.2] 'n_neighbors': [3, 5, 7], 'weights': ['uniform', 'distance'], 'metric': ['euclidean', 'manhattan'], 'algorithm': ['auto', 'ball_tree', 'kd_tree'], 'p': [1, 2] 'C': [0.1, 1, 5], 'gamma': ['scale', 'auto'], 'kernel': ['linear', 'poly', 'sigmoid'], 'degree': [3, 4] 'hidden_layer_sizes': [(50,), (100)], 'activation': ['tanh', 'logistic'], 'solver': ['adam', 'sgd'], 'alpha': [0.0001, 0.001], 'learning_rate': ['constant', 'adaptive'], 'learning_rate_init': [0.001, 0.01, 0.1]	'bootstrap': False, 'max_depth': 15, 'max_features': 'sqrt', 'min_samples_leaf': 2, 'min_samples_split': 2, 'n_estimators': 100
XGBoost Regressor		'colsample_bytree': 0.7, 'gamma': 0, 'learning_rate': 0.1, 'max_depth': 6, 'n_estimators': 200, 'subsample': 1.0
KNN Regressor		'algorithm': 'auto', 'metric': 'euclidean', 'n_neighbors': 3, 'p': 1, 'weights': 'distance'
SVR		'C': 5, 'degree': 3, 'gamma': 'scale', 'kernel': 'linear'
MLP Regressor		'activation': 'logistic', 'alpha': 0.001, 'hidden_layer_sizes': (100,), 'learning_rate': 'adaptive', 'learning_rate_init': 0.1, 'solver': 'adam'

Table 7
Model Evaluation Results (MAE, MSE, MedAE, R²).

	MAE	MSE	MedAE	R ²
Linear Regression	0.597	0.495	0.549	0.275
Decision Tree Regressor	0.097	0.106	0.000	0.843
Random Forest Regressor	0.077	0.037	0.023	0.945
XGBoost Regressor	0.067	0.021	0.007	0.968
KNN Regressor	0.178	0.174	0.000	0.744
SVR	0.541	0.632	0.250	0.075
MLP Regressor	0.280	0.144	0.218	0.788

(0.6322), and a very low R² (0.0755), indicating it was largely ineffective in capturing the underlying data structure. Linear Regression, though simple and interpretable, showed limited predictive capability with R² = 0.2754. This suggests that WQI relationships are likely non-linear and require more complex modeling approaches.

In the initial regression experiment, the SVR achieved a R² = 0.0755. Two primary factors contributed to this outcome. First, the regression target was potability (0/1/2), which is inherently an ordinal categorical label rather than a continuous variable; treating it as continuous is not well-suited to SVR and inevitably suppresses the R² score. Second, the initial SVR hyperparameter grid was restricted and did not include tuning of the RBF kernel or epsilon parameter, as only linear, polynomial, and sigmoid kernels were tested with limited values of C ∈ {0.1, 1, 5} and γ ∈ {scale, auto}. In follow-up runs, the hyperparameter search was expanded to include the RBF kernel, wider CCC and epsilon ranges, and log-scaled sweeps for γ. Despite these adjustments, SVR remained inferior to tree-based and boosting models, which consistently captured the nonlinear patterns in the dataset more effectively. These findings reinforce the classification results and highlight the limitations of SVR in this context. Importantly, the dataset used does not provide a continuous WQI field, making regression less appropriate.

Fig. 10 shows that ensemble models like Random Forest and XGBoost performed best, with predicted values closely matching the actual ones, while Linear Regression and Decision Tree were less accurate. Fig. 11 highlights that WQPs are the most important features, confirming their key role in predicting water quality.

Feature importance analysis revealed that parameters such as dissolved oxygen, BOD, EC, nitrate, and turbidity had the most significant influence on WQI classification and regression outcomes. Correlation matrices and heatmaps confirmed strong multicollinearity between certain variables, particularly between EC and TDS, and between DO and BOD, highlighting their critical roles in water quality dynamics. These findings collectively demonstrate that advanced ensemble machine learning models—particularly LightGBM and XGBoost—are highly effective in both classifying and predicting water quality using multivariate physicochemical data, offering practical tools for real-time water quality monitoring and environmental decision-making.

6. Discussion

Understanding the interrelationships among physicochemical parameters is crucial for assessing surface water potability and predicting the WQI. This study revealed that certain variables, such as DO, BOD, and phosphorus, showed moderate positive correlations with potability, while turbidity exhibited a notable negative correlation. The inverse association of turbidity aligns with established findings, where elevated turbidity has been shown to reduce light penetration, enhance microbial growth, and interfere with disinfection processes—making water less potable. The moderate positive correlation between BOD and nitrite, and between ammonia and phosphorus, suggests anthropogenic nutrient loading, potentially originating from domestic wastewater, agricultural runoff, and aquaculture activities (Hridoy et al., 2025b). High levels of BOD indicate the presence of biodegradable organic matter, which increases oxygen demand and leads to hypoxic conditions, thus impairing aquatic ecosystem health and potable water quality (Lv et al., 2024).

Boxplot analysis revealed that several water quality parameters, including temperature, calcium, alkalinity, and plankton, presented significant outliers. This heterogeneity could be attributed to localized variations in land use, industrial discharge, or seasonal hydrological changes—factors commonly reported in tropical river systems (Tanaka et al., 2021). Parameters such as ammonia, nitrite, and H₂S showed high dispersion, indicating sporadic contamination events, which may reflect either point-source discharges or temporary eutrophication episodes. These pollution spikes directly reduce the potability of water, especially in areas where water treatment infrastructure is limited or inconsistent. Density plot visualization by potability class emphasized distinct differences in water quality profiles. Excellent potability was characterized by low concentrations of BOD, CO₂, H₂S, ammonia, and nitrite, as well as

high DO and near-neutral pH levels—factors that are often considered critical indicators of drinking water safety (Dorpe, 2022). In contrast, poor potability coincided with elevated levels of pollutants and diminished oxygenation, consistent with the understanding that eutrophic or organically polluted waters often exhibit anoxic conditions unsuitable for human consumption (Akhtar et al., 2021).

Machine learning classification models yielded important insights. LightGBM emerged as the most accurate classifier, achieving high values across all performance metrics, including accuracy (0.9965), F1-score, and MCC. These findings align with previous studies showing that gradient boosting methods such as LightGBM and XGBoost offer superior generalization capabilities in environmental applications due to their handling of complex non-linear relationships and resistance to overfitting (Westhues et al., 2021). Ensemble methods such as Random Forest also performed well, highlighting the value of combining weak learners for robust water quality classification. By contrast, simpler models like Decision Tree and K-Nearest Neighbors exhibited reduced performance. AdaBoost showed the weakest predictive capability, consistent with its vulnerability to noisy datasets and sensitivity to class imbalance. SVM, while generally effective in high-dimensional spaces, may underperform when the data distribution is highly non-linear or sparse in certain class regions (Akusok and Eirola, 2019).

Although LightGBM's Accuracy, Precision, Recall, and F1 are numerically identical (0.9965). In a single-label multi-class setting, this is exactly what happens under micro-averaging: micro-precision = micro-recall = micro-F1 = accuracy. If that averaging scheme was used (as the values suggest), it should be stated explicitly. To provide class-wise nuance, we recommend also reporting macro (and/or weighted) F1, per-class metrics, and a confusion matrix, because micro-averages can mask minority-class errors. The strong SVM baseline (F1 = 0.9544; MCC = 0.9341) indicates that much of the signal is linearly separable after preprocessing; boosted trees add incremental gains via non-linear partitioning and better handling of feature interactions. KNN underperforms (F1 = 0.8531; MCC = 0.8124), consistent with sensitivity to scaling and the curse of dimensionality. Overall, the results are encouraging but warrant explicit leakage controls (pipeline-based cross-validation, group/time-aware splits) and ablation to confirm that improvements stem from model capacity rather than inadvertent information bleed.

For WQI regression, XGBoost outperformed all other models, achieving the lowest MAE (0.0677), MSE (0.0215), and the highest R² value (0.9685). Its ability to model non-linearities and incorporate regularization terms accounts for its exceptional accuracy (Wu et al., 2023). Random Forest also yielded robust results, albeit with slightly lower precision. Decision Tree and MLP models demonstrated moderate performance, likely due to their limited depth and optimization complexity. In contrast, linear models like SVR and Ordinary Least Squares regression performed poorly, underscoring the non-linear nature of relationships among water quality variables and the need for advanced algorithms in modeling such interactions (Masteali et al., 2025). Feature importance analysis revealed that DO, BOD, EC, nitrate, and turbidity were the most influential parameters in determining both potability classification and WQI scores. This aligns with previous studies that emphasize the significance of these variables as indicators of organic and nutrient pollution. Additionally, the strong multicollinearity between EC and TDS, and between DO and BOD, as visualized in the heatmap, further confirms the interdependent nature of water quality variables, highlighting the need for dimensionality reduction and feature selection during modeling.

The findings of this study reinforce the potential of machine learning-based frameworks as decision-support tools in water quality management. The ability of LightGBM and XGBoost to accurately classify potability and predict WQI demonstrates the utility of these models in real-time monitoring systems, especially in regions with limited laboratory capacity. Given the growing challenges of water pollution, urbanization, and climate-induced hydrological shifts, integrating data-

driven approaches into national water safety plans is essential for ensuring long-term sustainability and public health protection (Granata and Di Nunno, 2025).

To illustrate the practical operation of the alert mapping protocol, three representative records from the deployment system are presented. In the first case, a *Warning* was issued when the probability of poor water quality exceeded 0.80 despite adequate dissolved oxygen (e.g., ts = 2025-09-21 10:00, DO = 6.2 mg/L, WQI = 72, P(poor) = 1.0). The prescribed response included starting aeration, verifying sensor performance, and preparing for partial water replacement. In the second case, a *Critical* alert was triggered under combined conditions of high poor-class probability (≥ 0.90) and a low WQI (< 60) (e.g., ts = 2025-09-21 12:00, DO = 4.4 mg/L, WQI = 58, P(poor) = 1.0). Recommended interventions involved full aeration, partial water exchange, reduced feeding, spot-checking NH₃/NO₂⁻ levels, and reassessment at 10–15 min intervals. In the third case, a *Critical* alert was escalated independently of WQI or probability when dissolved oxygen fell below 3 mg/L (e.g., ts = 2025-09-21 13:00, DO = 2.8 mg/L, WQI = 55, P(poor) = 1.0), triggering the same intensive interventions. These examples demonstrate that the protocol effectively differentiates between Warning and Critical states while prescribing targeted, actionable responses tailored to the severity of water quality deterioration.

For real-time performance, these results show that vectorized inference amortizes fixed overhead, reducing per-sample latency substantially for modest batches (≤ 0.1 ms/sample at 64–512 records) (Table 8). Throughput peaks around ~18 k samples/s, with a slight plateau at the largest batch due to CPU/memory limits. Overall, the system comfortably meets “seconds-level” early-warning requirements for aquaculture monitoring and supports continuous streaming with ample headroom. The inference pipeline is stateless (no refitting) and uses the training-only scaler, preventing data leakage while enabling uninterrupted real-time scoring.

Importantly, the predictive insights gained through these models can be directly linked to early warning systems for aquatic disease outbreaks. Degraded water quality—characterized by low dissolved oxygen, elevated ammonia, and high turbidity—is often a precursor to disease emergence in aquaculture systems, such as bacterial infections (e.g., *Aeromonas*, *Vibrio* spp.), parasitic infestations, and viral epizootics. The ability of ML models to anticipate such water quality shifts with high precision enables timely interventions, reducing the likelihood of mass fish mortalities and minimizing economic losses. Furthermore, integrating real-time sensor data with predictive modeling enhances disease surveillance by identifying high-risk conditions before clinical signs appear in aquatic species. This ML-based framework not only strengthens ecosystem-based disease management but also aligns with One Health principles, recognizing the interconnection between environmental quality, aquatic animal health, and public health. Therefore, this study supports the development of automated, data-driven decision-support systems for sustainable aquaculture and water resource management in regions vulnerable to climate-induced water quality degradation and disease outbreaks. Future development of this early warning system will focus on establishing quantitative correlations between key water quality parameters (e.g., BOD, DO, temperature) and specific aquatic disease outbreaks, with defined threshold values. In addition, a technical framework will be designed to demonstrate how sensor data

Table 8
Inference latency and throughput (CPU).

Batch size	Latency (s) mean $\pm$ SD	Approx. throughput (samples)	Approx. ms/sample
1	0.00460 $\pm$ 0.00032	217	4.60
16	0.00500 $\pm$ 0.00030	3200	0.31
64	0.00600 $\pm$ 0.00010	10,667	0.094
128	0.00820 $\pm$ 0.00020	15,610	0.064
256	0.01440 $\pm$ 0.00040	17,778	0.056
512	0.03280 $\pm$ 0.00450	15,610	0.064

can be integrated into real-time predictive models and automated alert systems for farmers.

This study underscores the interrelationships among physicochemical parameters and their roles in assessing surface water potability and predicting the WQI. Turbidity exhibited a negative correlation with potability, while parameters such as DO, BOD, and phosphorus showed moderate positive associations. Among machine learning models, LightGBM and XGBoost achieved superior accuracy in both potability classification and WQI prediction due to their robustness in modeling complex non-linear interactions. Feature importance analysis consistently identified DO, BOD, EC, nitrate, and turbidity as key indicators of water quality. These findings confirm the reproducibility and reliability of data-driven models in real-time water quality monitoring, reinforcing their potential for integration into national water safety management systems.

7. Conclusion

This study confirms that advanced ensemble machine learning models, particularly LightGBM and XGBoost, deliver highly accurate and robust predictions for both water quality classification and WQI estimation. Among these, LightGBM exhibited the highest predictive accuracy, demonstrating superior capability in handling complex, non-linear relationships in multivariate physicochemical datasets. Key parameters—including DO, BOD, EC, nitrate, and turbidity—were identified as critical contributors to water quality prediction. The observed interrelationships among these variables underscore the necessity of incorporating feature interactions into model development. Overall, the findings highlight the practical utility of these machine learning approaches for real-time water quality monitoring, offering valuable support for evidence-based environmental management and sustainable water resource planning, while enhancing public health protection.

CRediT authorship contribution statement

Md. Abdullah Al Mamun Hridoy: Writing – original draft, Visualization, Validation, Software, Formal analysis, Data curation, Conceptualization. **Abdullah Ibna Shawkat:** Data curation. **Chiara Bordin:** Data curation, Conceptualization. **Mahima Ranjan Acharjee:** Writing – review & editing. **Andleeb Masood:** Writing – review & editing, Validation. **Azeez Olalekan Baki:** Writing – review & editing, Validation. **Md. Abdullah Al Mamun:** Writing – review & editing, Validation.

Clinical trial number

Internet of Things (IoT) devices and subsequent analysis using deep learning techniques. As such, clinical trial procedures do not apply to our work.

Consent to publish declaration

Not applicable

Ethics and consent to participate declarations

Not applicable.

Guarantor

No persons or third-party services were involved in the research or manuscript preparation who are not listed as an author or acknowledged in the manuscript.

Research registration number

All code used for data analysis and model development is publicly available at: <https://github.com/MHridoy-123/Water-Quality-prediction-model>

Declaration of Generative AI and AI-assisted technologies in the writing process

Artificial intelligence (AI) software, including machine learning libraries such as TensorFlow, was utilized for data analysis and modeling in this study. While no AI tools were used to write or prepare the manuscript text, AI-assisted tools were employed solely to support ideation and conceptual thinking during the research process.

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Declaration of competing interest

The authors declare that there are no conflicts of interest. No funding, commercial, or institutional pressures influenced the design, execution, or reporting of this study. All affiliations listed were included solely for administrative and identification purposes.

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Data availability

<https://data.mendeley.com/datasets/y78ty2g293/1>

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